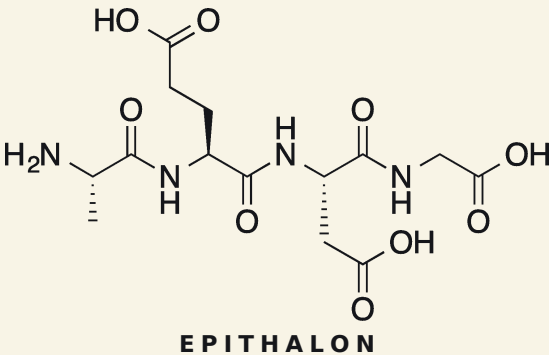


The Peptide Pocket Guide

A bench-side reference to seventy-two peptide therapeutics, biologics, and mixtures, with *GRADE evidence ratings*, mechanism, dosing, and regulatory status for each.



FADY HANNAH-SHMOUNI, MD FRCPC

Clinical Endocrinologist & Geneticist
University of British Columbia
Founder & CEO, Hormonally.ai

2026 EDITION

The Peptide Pocket Guide

Peptide therapeutics, biologics, and mixtures, rated by the strength of the evidence behind them.

A compact reference covering the major regulated and unregulated peptides encountered in wellness, longevity, and endocrine practice.

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How to use this guide

What each entry contains, how the book is ordered, and what the shaded box on every page is for.

This pocket guide covers seventy-two entries — peptide therapeutics, peptide-derived biologics, peptide mixtures, and peptide-adjacent comparators (notably small-molecule oral GLP-1 receptor agonists). FDA- and EMA-approved drugs sit alongside the investigational and compounding-accessible agents commonly encountered in modern practice.

Each entry occupies a single page and follows the same template: a pharmacologic class, the canonical name and any brand names, history where a documented development record exists, sequence, mechanism, indications, dosing, evidence, safety, adverse reactions, a clinical caveat, and regulatory status. History appears where a development record adds something a reader would use; six entries omit it.

READING AN ENTRY

GRADE letter

A through D, the strength of the clinical evidence for the listed use. It rates evidence, not legality — an approved drug can carry a low GRADE for the use it is sold for.

Class label

The pharmacologic class, at the top right of the page.

Regulatory status

What the molecule actually is in law: approved, investigational, compounding-accessible, or research-only.

Clinical caveat

The shaded box. The page's operative warning.

HOW THE BOOK IS ORGANIZED

Molecules are grouped by therapeutic class — metabolic, growth-hormone axis, regenerative, dermatology, bone, somatostatin, melanocortin, reproductive, mitochondrial and longevity, neuropeptides, and gut and antimicrobial — and ordered most-evidence-rich first within each class. The back matter holds clinical pearls, a lab and monitoring reference, the regulatory and medicolegal chapter, and the bibliography.

This guide is a clinical reference, not a recommendation. Inclusion does not imply that a molecule is approved, appropriate for any patient, or standard of care.

How strong is the evidence?

The GRADE letter on every entry, what it rates, and why it is not the same scale as the companion volume's.

Every entry carries a GRADE letter. GRADE reflects the strength of the clinical evidence for the listed use — not mechanistic plausibility, popularity, compounding availability, or clinical fashion.

GRADE A	High-quality evidence. Multiple adequately powered randomized trials, or one large trial with consistent supporting data.
GRADE B	Moderate. Randomized evidence with limitations, or consistent observational data in humans.
GRADE C	Low. Small or uncontrolled human studies, surrogate endpoints, or conflicting results.
GRADE D	Very low or absent human evidence. Animal, in vitro, or mechanistic reasoning only.

A low GRADE is not a verdict that a molecule does not work. It states that the published evidence cannot yet support a confident claim that it does.

A SEPARATE SCALE FROM THE COMPANION VOLUME

The Expert Witness Pocket Guide rates how settled the law is on a question, using the letters A through D for a different purpose entirely. A molecule can rest on settled law and weak evidence, or on strong evidence and unsettled law. Do not read a GRADE letter here as an authority rating there.

WHY RESEARCH-ONLY MOLECULES ARE INCLUDED

This guide covers approved drugs, investigational peptides in active trials, compounded agents, and research-use-only substances encountered in practice. Including a gray-market peptide does not endorse its use. Patients arrive already taking BPC-157, TB-500, and Vilon; understanding the preclinical evidence, the known harms, and the evidentiary gaps is part of treating them.

Where the GRADE ratings come from

The rating method carried forward from the first edition.

Every entry carries a GRADE letter rating derived from the Hormonal.ai evidence database, which adapts the standard GRADE framework to peptide pharmacology.

GRADE A	High-quality evidence. Multiple pivotal RCTs with consistent findings; broadly accepted standard of care for the listed indications.
GRADE B	Moderate-quality evidence. Some RCTs and substantial observational data; conclusions reasonably stable but not yet definitive.
GRADE C	Low-quality evidence. Limited human data, mostly observational or small trials; mechanistic and animal data may be strong but human translation incomplete.
GRADE D	Very low-quality evidence. Mostly preclinical, anecdotal, or industry-driven claims; clinical use carries meaningful uncertainty.

A GRADE C or D rating does not mean a peptide is ineffective -- only that the evidence supporting its use does not yet meet the standard of approved therapeutics.

What the references mean

Every reference in this edition was checked against PubMed; non-indexed sources are explicitly flagged. This is what that found, and what the flags mean.

Every reference in this edition was checked against PubMed; non-indexed sources are explicitly flagged. Each printed PMID was retrieved and its title, first author, journal, volume, pages, and year compared against the citation as printed.

That check changed a great deal. Of 151 references carried forward from the first edition, 51 were exactly right; 19 carried a wrong field; 31 resolved to an entirely different paper; and 13 that carried no identifier were matched to one. All of those are corrected here.

33 of the 171 references in this edition are not indexed in PubMed at all — manufacturer press releases, FDA labels, trial registry records, a 1922 paper, and Russian-language literature. These print with a flag rather than being dropped, because the claim they support is still in the book and a reader is entitled to know the basis is not a peer-reviewed indexed article.

A reference marked [not indexed] is not a discredited source. It is a source a reader cannot check on PubMed, which is a different and useful fact.

Editorial standards and evidence philosophy

What inclusion means, what it does not mean, and how this guide treats thin evidence.

This guide is a clinical reference, not a recommendation or endorsement. Inclusion does not imply that a peptide is approved, recommended for practice, appropriate for any individual patient, or standard-of-care.

EVIDENCE-FIRST, BY GRADE

Every entry is rated A-D by GRADE methodology, which reflects the strength of clinical evidence -- not molecular plausibility, popularity, compounding availability, or fashion. An FDA-approved drug may carry a lower grade for a particular off-label use than an investigational compound supported by stronger human evidence; regulatory status and evidentiary certainty are separate axes. Mechanistic appeal alone never raises a grade.

GRADE A does not mean "use this"; GRADE D does not mean "do not." Clinical decisions require GRADE plus regulatory status, contraindications, patient preference, and judgment.

WHY INVESTIGATIONAL AND GRAY-MARKET COMPOUNDS ARE INCLUDED

The guide spans approved drugs, investigational peptides in active trials, 503A-compounded agents, and research-only substances seen in gray-market settings. Inclusion is not endorsement -- it reflects clinical reality: agents like BPC-157, TB-500, and Dihexa are used off-label without approval, and clinicians need their preclinical evidence, known harms, and evidentiary gaps to manage patients who use them.

FOUR TIERS, KEPT SEPARATE

Each entry separates what is known from what is hoped, in plain language:

EDITORIAL STANDARDS AND EVIDENCE PHILOSOPHY · CONTINUED

- Established evidence -- FDA approval, positive Phase 3 RCTs, meta-analysis consensus.
- Emerging evidence -- Phase 2 human data, small observational studies, mechanism validated in humans.
- Clinical culture -- usage in anti-aging clinics, bodybuilding, and biohacker communities; documented, never conflated with evidence.
- Mechanistic plausibility -- in-vitro activity and animal models; not clinical evidence until translated to human efficacy and safety.

A peptide supported by animal or preclinical data only has no human efficacy evidence, regardless of GRADE rating or mechanistic appeal.

CONTEXT WITHOUT ENDORSEMENT

Descriptions like "widely used in anti-aging clinics" or "common microdosing protocol" are observations, not endorsements.

For Grade C/D, "Research only," "Not approved," or "Gray market" entries, the clinical caveat states plainly whether use carries unjustifiable risk or no human efficacy trials support this use — factual statements, not hedges.

A clinician meeting a patient on an off-label peptide should leave each entry knowing what the culture believes, what the evidence shows, and where the gaps are.

Peptide fundamentals

Definitions, classification, manufacturing, and the regulatory grades a single molecule can occupy.

Definitions, classification, manufacturing, regulatory grades, and how to read a Certificate of Analysis — the foundation behind every entry in this guide.

WHAT IS A PEPTIDE?

A peptide is a short chain of amino acids linked by peptide bonds. The biochemical distinction from a protein is conventionally one of length: peptides are typically 2–50 residues; proteins are longer, fold into stable tertiary structures, and usually exceed 10 kDa.

The boundary is genuinely fuzzy — insulin (51 residues, two disulfide-linked chains) crosses the threshold by one residue and is variably classified as a polypeptide or a small protein. We list it as a 51-residue exception under the polypeptide class.

THE FDA'S REGULATORY LINE

The biochemical definition above is descriptive; the FDA's definition is operational and consequential. Under the BPCIA (2009) as amended by the Further Consolidated Appropriations Act (2020), the agency draws the peptide/protein line at 40 alpha amino acids:

≤ 40 alpha amino acids	Peptide, regulated as a drug under the FD&C Act (NDA pathway)
41–99 alpha amino acids, chemically synthesized	"Chemically synthesized polypeptide" — still regulated as a drug under the FD&C Act
> 40 alpha amino acids, not chemically synthesized (recombinant or extracted)	Protein, regulated as a biological product under the PHS Act (BLA pathway)
≥ 100 alpha amino acids	Always a protein/biologic regardless of synthesis route

The classification matters: biologics face additional statutory protections (12-year exclusivity, biosimilar pathway), more rigorous CMC review, and are largely ineligible for 503A pharmacy compounding. Drugs follow the smaller-molecule NDA pathway with shorter exclusivity and a more permissive compounding regime.

PEPTIDE FUNDAMENTALS • CONTINUED

CLASSIFICATION BY CHAIN LENGTH

Dipeptides & tripeptides	2–3 residues. Examples: GHK (3), KPV (3)
Oligopeptides	4–20 residues. Examples: Oxytocin (9), GnRH (10), Octreotide (8), Kisspeptin-10 (10), BPC-157 (15)
Polypeptides	20–50 residues. Examples: GLP-1 analogues (30–31), Glucagon (29), Calcitonin (32), Tesamorelin (44)
Small proteins	≥51 residues. Examples: Insulin (51, boundary case), GH (191), Follistatin (315), Cerebrolysin (mixture)

WHY CHAIN LENGTH MATTERS

Residue count drives pharmacology and, near 40 residues, regulatory status:

≤ 10 residues	Solid-phase synthesis. Half-life often minutes (oxytocin ≈ 3 min); rapidly cleared by peptidases, needing frequent or sustained-release dosing.
10–30 residues	Solid-phase synthesis still cost-effective. Half-life extended by D-amino-acid substitution (octreotide), N-methylation, cyclization, or PEGylation.
> 30 residues	Recombinant production becomes attractive. Half-life extended by fatty-acid acylation (semaglutide $t_{1/2}$ ~7 days), Fc-fusion (dulaglutide), or depot (lanreotide).

A NOTE ON BIOREGULATORS

One functional class cuts across chain length: the bioregulators (Khavinson "cytomedines") – very short di-/tripeptides claimed to regulate tissue-specific genes. Examples here include Epitalon and Pinealon (with Cortagen and Vesugen). Defined by mechanism rather than structure, the category rests on one Russian group without independent Western replication.

The bioregulator category rests on one Russian research group without independent Western replication – as the GRADE C/D ratings reflect.

Route selection is determined by molecular weight, gut stability, target tissue, and the half-life the formulation needs to achieve.

PEPTIDE FUNDAMENTALS • CONTINUED

Subcutaneous (SC) — ~70–90% bioavailability	The default for most peptide drugs. Self-administered with a 29–31 gauge insulin syringe; rotate injection sites to avoid lipohypertrophy.
Intramuscular (IM) — ~75–95% bioavailability	Faster onset than SC, more painful. Used for depot formulations (leuprolide, lanreotide), emergency glucagon, and a few clinic-administered peptides.
Intravenous (IV) — 100% bioavailability	Clinical setting only — insulin for DKA, glucagon for hypoglycemia, octreotide infusions for variceal bleeding.
Oral — <1% native bioavailability	Most peptides are destroyed in the gut. Successful oral peptides either use absorption enhancers (SNAC for oral semaglutide) or are not meant to be absorbed (linaclotide acts in the gut lumen).
Intranasal — ~5–30% bioavailability	Useful for calcitonin, desmopressin, oxytocin, Selank, Semax. Direct CNS access via the olfactory nerve makes intranasal attractive for neuropeptides.
Sublingual / buccal — variable bioavailability	Bypasses first-pass metabolism. Limited by peptide molecular weight and lipophilicity; mostly research-grade preparations.
Topical — local only	Skin barrier limits systemic absorption. GHK-Cu cosmetics, BPC-157 ointments. Local effects only — claims of systemic activity from topical application are usually unsupported.

RECONSTITUTION

Most injectable peptides ship as lyophilized (freeze-dried) powder that must be reconstituted before use. The mechanics matter: forceful reconstitution shears and denatures peptide, and repeated freeze-thaw and agitation are recognised degradation routes. Published loss figures are formulation-specific, so no single percentage applies across peptides.

PEPTIDE FUNDAMENTALS • CONTINUED

Solvent	Bacteriostatic water for injection (BWFI, 0.9% benzyl alcohol) for multi-dose vials; sterile water for single-use only.
Technique	Inject solvent slowly down the inside wall of the vial. Do not aim the stream at the powder. Swirl gently — do not shake. Shaking creates foam and can denature the peptide.
Concentration	Pick a reconstitution volume that gives convenient dosing. A 5 mg vial reconstituted in 2.5 mL BWFI yields 2 mg/mL; a 250 µg dose is then 0.125 mL, or 12.5 units on a U-100 insulin syringe.

Pre-dose inspection: the reconstituted solution should be clear and colourless. Any cloudiness, particulate matter, or yellowing indicates the peptide has degraded — discard and start a new vial.

STORAGE RULES

Lyophilized vial	Refrigerate at 2–8°C. Frozen (–20°C) for long-term storage. Many peptides tolerate room temperature for weeks if dry and well-sealed.
Reconstituted vial (BWFI)	Refrigerate at 2–8°C. Benzyl alcohol is bacteriostatic but does not prevent chemical degradation. Use the in-use period the manufacturer or compounding pharmacy assigns; where a product has none, a pharmacy-assigned beyond-use date governs rather than a generic figure.
Reconstituted vial (sterile water)	No preservative against microbial growth, so the in-use period is short and is set by the assigned beyond-use date for the preparation, not by a rule of thumb.

- Light and heat: store in original container, away from direct light. Avoid leaving in a hot car, a window, or near radiators.
- Cold-chain shipping with ice packs protects the molecule; ambient shipping exposes it to temperatures that drive degradation, with the extent depending on the peptide and the excursion.
- Travel: transfer vials to an insulated bag with refrigerated ice packs. Tolerance of an ambient excursion depends on the formulation and on the temperature reached, so record the excursion and check it against the product information rather than assuming a fixed allowance.

PEPTIDE FUNDAMENTALS • CONTINUED

Peptides are intrinsically unstable molecules. Knowing the degradation pathways tells you how to store, when to discard, and what to look for in the vial.

THE FIVE DEGRADATION PATHWAYS

Oxidation	Of methionine, cysteine, and tryptophan residues. Mitigated by antioxidants and oxygen-free packaging.
Hydrolysis	Of peptide bonds in solution — accelerated by extreme pH, heat, and time. Lyophilized form is dramatically more stable.
Deamidation	Of asparagine and glutamine to acidic forms — a major cause of slow potency loss in stored solutions, occurring even refrigerated.
Aggregation	Peptides dimerize, oligomerize, or precipitate at high concentration. Surfactants and gentle handling help.
Adsorption	Peptides stick to glass and plastic surfaces, silently depleting low-concentration solutions over weeks.

VISUAL SIGNS OF DEGRADED PRODUCT

- Cloudy or particulate solution after reconstitution
- Yellowing of the powder or solution
- Loss of clinical effect at a previously effective dose
- Precipitate that fails to redissolve on gentle warming
- Off-smell from the vial (rare but a hard signal)

PREFILLED PEN STABILITY

Prefilled pens are factory-formulated with preservatives and tested under cGMP. The manufacturer-validated in-use stability of the major GLP-1 pens, per FDA-approved labelling:

Ozempic and Wegovy (semaglutide)	56 days at controlled room temperature (15–30°C / 59–86°F) after first use; refrigeration permitted but does not extend the window.
Mounjaro KwikPen (tirzepatide, multi-dose)	30 days at room temperature once in use.
Single-dose Mounjaro and Zepbound pens	21-day room-temperature limit before use, then single injection and disposal.
Compounded semaglutide and tirzepatide	Pharmacy-assigned beyond-use dates (BUDs) governed by USP <797>, typically shorter than the labelled in-use windows for branded pens.

PEPTIDE FUNDAMENTALS • CONTINUED

Gray-market pens and vials are sold without manufacturer stability validation, cGMP oversight, or independent testing. Stability is undocumented and unverifiable; the most conservative of the above limits should be presumed at best.

SOLID-PHASE PEPTIDE SYNTHESIS (SPPS)

Invented by Bruce Merrifield in 1963 (Nobel Prize 1984). The peptide is built one amino acid at a time on a polymer resin.

Each cycle deprotects the chain, couples the next Fmoc-protected amino acid, and washes — typically 30-60 minutes per residue.

After assembly, the peptide is cleaved from resin, side-chains deprotected, and the crude product purified by reverse-phase HPLC.

Yields fall geometrically: 98% per coupling step x 50 cycles ≈ 36% overall yield.

The practical ceiling for routine pharmaceutical manufacture is ~50 residues. Beyond that, recombinant approaches become more economical.

- Resin bead + AA1
- + AA2
- + AA3 ...
- Cleave
- Crude peptide
- HPLC >= 98% pure

HALF-LIFE EXTENSION STRATEGIES**RECOMBINANT DNA TECHNOLOGY**

For longer peptides and proteins. The peptide-coding DNA is inserted into a host organism — *E. coli* (insulin), *Pichia pastoris* (some calcitonin), or mammalian CHO cells (Fc-fusion drugs).

Used for insulin, growth hormone, and most GLP-1 analogues; the recombinant backbone is then chemically conjugated with fatty acids or other half-life extenders.

Fatty-acid acylation	Semaglutide (C18 diacid) — $t_{1/2}$ ~7 days
Fc fusion	Dulaglutide (IgG4 Fc) — $t_{1/2}$ ~5 days
Depot formulation	Lanreotide autogel — $t_{1/2}$ ~28 days

PEPTIDE FUNDAMENTALS • CONTINUED

The same molecule can exist in five radically different regulatory states. Understanding the hierarchy is essential before prescribing or recommending any peptide.

FDA

FDA-approved (or equivalent). An effective approved application, with a labeled indication, dosing, contraindications, and safety information. Approval may be traditional or accelerated: an accelerated approval can rest on an intermediate or surrogate endpoint in a small trial, so approval alone does not imply a completed Phase 3 program. Manufactured to cGMP with post-market surveillance. Coverage and reimbursement are separate questions from approval.

- Examples: insulin, semaglutide, tesamorelin, leuprolide, octreotide.
- About half the peptides in this guide fall into this category.

503B

503B outsourcing facility. Compounded under cGMP for office-use stock. Registered with the FDA and inspected. The compound itself is not FDA-approved — only the manufacturing standard is. Used for shortages, off-label needs, or patient-specific requirements. Quality bar is high; documentation is comprehensive.

503A

503A compounding pharmacy. State-licensed pharmacy compounding against a specific patient prescription; a different regulatory bar from 503B rather than simply a lower one. The bulk substance must satisfy a statutory pathway: the 503A Bulks List at 21 C.F.R. 216.23, an applicable USP or NF monograph, or a component of an FDA-approved drug.

In September 2023 FDA's 503A Category 2 list identified several peptides — BPC-157, TB-500, melanotan II among them — as raising significant safety risks for compounding. That roster has since changed: as of 14 May 2026 Category 2 holds six substances, the peptide-class entries being ibutamoren mesylate and kisspeptin-10, and most of the 2023 peptides appear on FDA's table of withdrawn nominations rather than in any category. Category status is an interim enforcement posture, not the statutory Bulks List at 21 C.F.R. 216.23.

PEPTIDE FUNDAMENTALS • CONTINUED

RUO

Research Use Only. Sold by chemical-supply companies (Bachem, Sigma, Anaspec, others) labeled "for research only — not for human consumption." Purity varies from pharmaceutical-grade (Bachem) to unverified (smaller suppliers). Outside pharmaceutical regulation.

GRAY

Gray market / underground. "Peptide research" websites with deliberate ambiguity. Quality ranges from acceptable to dangerous. May contain wrong peptides, sub-potent material, bacterial endotoxin, residual TFA, or heavy metals. No regulatory oversight.

- Purchased material has failed independent analysis: three semaglutide vials bought from illegal online sellers contained 28.6-38.7% more peptide than labelled, at 7.7-14.4% purity against a 99% claim (Ashraf et al. J Med Internet Res 2024;26:e65440 (PMID 39509151)).

OVERSIGHT HIERARCHY

- FDA approved
- 503B outsourcing facility (cGMP)
- 503A compounding pharmacy
- Research Use Only (RUO)
- Gray market / underground

The RUO disclaimer provides no legal protection — the FDA regulates based on inferred intended use, not the label on the product.

A COA is the supplier's own quality attestation, not an independent certification. A serious COA tells you what the supplier found on a named lot: whether the mass matches the expected molecule, how pure the material was by HPLC, what counter-ion it carries, and what the endotoxin and microbial results were. What it cannot establish on its own is that a product is safe to inject — that turns on sterility, the dose and route, the conditions the material was held in since testing, and whether the lot tested is the lot in hand.

IDENTITY

Mass spectrometry confirms the molecular weight matches the expected peptide. The theoretical monoisotopic mass should be reported alongside the observed mass. Deviations > 1 Da indicate sequence problems.

PEPTIDE FUNDAMENTALS • CONTINUED

PURITY

HPLC chromatogram with retention time and peak integration.

Pharmaceutical grade commonly $\geq 98\%$ by HPLC — a purchasing convention, not a statutory COA requirement

RUO typically 95–99%

Below 95% significant impurity load

WATER CONTENT

Karl Fischer titration. Lyophilized peptides typically retain 3–10% residual moisture. Higher water content reduces stability.

COUNTER-ION (TFA / ACETATE)

Synthetic peptides ship as TFA or acetate salts. A 5 mg "peptide" vial may contain 4.2 mg net peptide after subtracting the counter-ion. Pharmaceutical preparations always report this.

ENDOTOXIN (LAL)

Endotoxin is measured by the bacterial endotoxins test (Limulus amebocyte lysate). There is no single universal per-milligram limit: the allowable endotoxin for a parenteral product is derived from the maximum human dose and the route, so the same concentration can pass for one product and fail for another. A COA reporting endotoxin without stating the dose and route it was judged against has not established suitability for injection.

RUO peptides frequently have no endotoxin testing — a hard contraindication to injection.

MICROBIAL CONTAMINATION

Total aerobic count, yeast and mold count, absence of pathogens. Critical for any injectable preparation.

HEAVY METALS

Lead, cadmium, mercury, arsenic by ICP-MS. Within ICH Q3D limits.

RESIDUAL SOLVENTS

DMF, DMSO, methanol from SPPS. Within ICH Q3C limits.

PEPTIDE FUNDAMENTALS · CONTINUED**WHAT YOU SHOULD NOT SEE ON A SERIOUS COA**

- A bare claim of "purity > 99%" with no chromatogram or supporting data.
- No mass spectrum, or a spectrum without a labelled expected mass.
- No batch or lot number; no manufacture date.
- A COA dated more than 12 months before purchase — peptides degrade in storage.
- Generic identical-looking COAs across different products from the same vendor.
- No endotoxin or microbial data on a product marketed for injection.
- A COA referring to a "typical" batch rather than the specific batch shipped.

PRACTICAL HEURISTIC

A lot-specific COA supplied with the vial, naming the batch shipped, is the minimum documentation to look for. Responsiveness to a documentation request is a commercial signal rather than a quality standard, but an unanswered request leaves the material undocumented.

If you must request a COA and the response is delayed, generic, or never arrives, the product is probably not what it claims to be — and certainly not safe to inject.

PEPTIDE FUNDAMENTALS • CONTINUED**FOR 503A/503B COMPOUNDED PEPTIDES**

Compounding pharmacies should provide a COA documenting identity, purity, sterility, and endotoxin. Ask which testing laboratory ran the analysis, the date of testing, the USP-797 / 800 compliance status of the facility, and the lot number that links the COA to the vial in hand.

A pharmacy that cannot produce this documentation on request is not operating to the standards their license requires.

FOR RESEARCH-USE-ONLY PEPTIDES

If the product is being used outside its labelled purpose, the standard of evidence shifts to the user. At minimum:

- Confirm identity by independent mass spectrometry.
- Confirm purity by independent HPLC.
- Run an endotoxin test before any injectable use.
- Document the source, lot, and date for every dose.

None of this turns RUO into pharmaceutical grade, but it surfaces problems before they reach a patient.

Contents

How to use this guide	1
How strong is the evidence?	2
Where the GRADE ratings come from	3
What the references mean	4
Editorial standards and evidence philosophy	5
Peptide fundamentals	7
METABOLIC & INCRETIN	21
Dulaglutide	GRADE A 22
Exenatide	GRADE A 23
Glucagon	GRADE A 24
Insulin	GRADE A 25
Liraglutide	GRADE A 26
Orforglipron	GRADE A 27
Retatrutide	GRADE A 28
Semaglutide	GRADE A 29
Tirzepatide	GRADE A 30
Pramlintide	GRADE B 31
GROWTH-HORMONE AXIS	32
Somatropin (hGH)	GRADE A 33
Tesamorelin	GRADE A 34
Sermorelin	GRADE B 35
AOD-9604	GRADE C 36
CJC-1295	GRADE C 37
GHRP-2	GRADE C 38
Ipamorelin	GRADE C 39
GHRP-6	GRADE D 40
Hexarelin	GRADE D 41
IGF-1 LR3	GRADE D 42
REGENERATIVE & REPAIR	43
ARA-290	GRADE B 44
Thymosin α 1	GRADE B 45
BPC-157	GRADE C 46
Follistatin	GRADE C 47
KPV	GRADE C 48
TB-500 (Ac-LKKTETQ)	GRADE D 49
Thymulin	GRADE D 50

DERMATOLOGY & COSMETIC	51
GHK	GRADE C 52
GHK-Cu	GRADE C 53
Argireline	GRADE D 54
Leuphasyl	GRADE D 55
Matrixyl 3000	GRADE D 56
PTD-DBM	GRADE D 57
Palmitoyl Tripeptide-1	GRADE D 58
Zinc Thymulin	GRADE D 59
BONE METABOLISM	60
Abaloparatide	GRADE A 61
Calcitonin (salmon)	GRADE A 62
Teriparatide	GRADE A 63
SOMATOSTATIN ANALOGUES	64
Lanreotide	GRADE A 65
Octreotide	GRADE A 66
Pasireotide	GRADE A 67
MELANOCORTIN & SEXUAL	68
Afamelanotide	GRADE A 69
Bremelanotide	GRADE A 70
Melanotan II	GRADE C 71
PT-141	GRADE C 72
REPRODUCTIVE & PITUITARY	73
Desmopressin	GRADE A 74
Goserelin	GRADE A 75
Leuprolide	GRADE A 76
Oxytocin	GRADE A 77
hCG	GRADE A 78
Kisspeptin-10	GRADE B 79
MITOCHONDRIAL & LONGEVITY	80
SS-31 (Elamipretide)	GRADE B 81
Epitalon	GRADE C 82
Klotho	GRADE C 83
MOTS-c	GRADE C 84
Vilon	GRADE C 85
5-Amino-1MQ	GRADE D 86
FOXO4-DRI	GRADE D 87

G-Klotho Frag 33	GRADE D	88
Humanin	GRADE D	89
NEUROLOGICAL		90
Cerebrolysin	GRADE B	91
Pinealon	GRADE C	92
Selank	GRADE C	93
Semax	GRADE C	94
Dihexa	GRADE D	95
DSIP (Delta Sleep-Inducing Peptide)	GRADE D	96
PE-22-28	GRADE D	97
INVESTIGATIONAL ONCOLOGY		98
Met-5-Enkephalin	GRADE D	99
PNC-27	GRADE D	100
GI & OTHER		101
Linacotide	GRADE A	102
LL-37 (Cathelicidin)	GRADE C	103
Larazotide acetate	GRADE C	104

PART

Metabolic & Incretin

Incretin agonists and insulin – the most heavily evidenced peptides in the book, and the ones most often compounded outside their approvals.

IN THIS PART

- Dulaglutide
- Exenatide
- Glucagon
- Insulin
- Liraglutide
- Orforglipron
- Retatrutide
- Semaglutide
- Tirzepatide
- Pramlintide

Dulaglutide

Trulicity

HISTORY

Eli Lilly's once-weekly GLP-1 agonist; FDA-approved 2014. The first GLP-1 to use an Fc-fusion strategy for prolonged half-life. Now largely overshadowed by tirzepatide from the same manufacturer.

SEQUENCE GLP-1(7-37) analogue fused to human IgG4 Fc fragment
Fc fusion (~120 kDa)

MECHANISM

GLP-1 receptor agonist; IgG4 Fc fragment extends half-life to ~5 days via FcRn recycling.

INDICATIONS

- Type 2 diabetes
- MACE reduction in T2DM ± established CVD

DOSING

SC weekly 0.75 mg → 1.5 mg → up to 4.5 mg

EVIDENCE · GRADE A RATES TYPE 2 DIABETES AND MACE REDUCTION
REWIND trial: significant MACE reduction in 9,901 patients with T2DM at lower CV risk than prior trials.

SAFETY

GI side effects · Pancreatitis risk · MTC/MEN2 contraindication

ADVERSE REACTIONS

Most common: nausea (12–21%), diarrhea (12–13%), vomiting (6–13%), abdominal pain, decreased appetite, dyspepsia, fatigue. Less common: pancreatitis, hypersensitivity reactions, hypoglycemia (with insulin/SU).

Clinical caveat. Single-use pen makes administration easier than semaglutide for elderly or dexterity-limited patients. REWIND demonstrated CV benefit in lower-risk T2DM than other GLP-1 outcome trials. Hypothesis-generating signal: Probst et al. (JCI Insight 2023) reported reduced alcohol consumption during smoking cessation; preclinical signal for AUD — under active investigation, not clinical consensus.

REGULATORY STATUS FDA-approved (Trulicity, 2014) · EMA

REFERENCES

- Gerstein HC et al. REWIND. Lancet 2019;394:121 (PMID 31189511)
Probst et al. JCI Insight 2023;8: (PMID 37991022)

Exenatide

Byetta · Bydureon

HISTORY

Derived from exendin-4, a peptide isolated from Gila monster venom. The first GLP-1 receptor agonist on the market (FDA 2005). Largely displaced by once-weekly agents.

SEQUENCE HEGTFTSDLSKQMEEEAVRLFIEWLKNNGPSSGAPPPS

39 residues

MECHANISM

GLP-1 receptor agonist; resistant to DPP-4 cleavage due to alanine-to-glycine substitution at position 2.

INDICATIONS

— Type 2 diabetes (adjunct)

DOSING

SC twice daily (Byetta) 5 µg → 10 µg twice daily

SC weekly (Bydureon) 2 mg once weekly

EVIDENCE

EXSCEL trial showed neutral cardiovascular outcomes in T2DM. Efficacy generally lower than semaglutide/tirzepatide.

SAFETY

GI side effects · Injection-site nodules with Bydureon (depot) · Pancreatitis risk · MTC contraindication

ADVERSE REACTIONS

Most common: nausea (44%), vomiting (13%), diarrhea (13%), feeling jittery, dizziness, headache, dyspepsia. Bydureon depot: injection-site nodules (long-lasting), pruritus. Less common: acute pancreatitis, AKI.

Clinical caveat. Twice-daily Byetta is now rarely used; weekly Bydureon depot creates persistent injection-site nodules in roughly 17% of patients. EXSCEL trial was CV-neutral, unlike semaglutide and liraglutide. Hypothesis-generating signal: one positive RCT for smoking cessation with nicotine patch; preclinical signal for AUD via dopaminergic-reward modulation — under active investigation, not clinical consensus.

REGULATORY STATUS FDA-approved (Byetta 2005, Bydureon 2012) · EMA

REFERENCES

Buse et al. Diabetes Care 2004;27:2628-35 (PMID 15504997)

Holman RR et al. EXSCEL. NEJM 2017;377:1228 (PMID 28910237)

Glucagon

GlucaGen · Baqsimi · Gvoke

HISTORY

Identified in 1923 as a contaminant of early insulin preparations causing hyperglycemia. Used for emergency hypoglycemia treatment for decades; nasal (Baqsimi 2019) and pre-filled pen (Gvoke 2019) formulations are recent advances.

SEQUENCE HSQGTFTSDYSKYLSRRAQDFVQWLMNT

29 residues

MECHANISM

Activates the hepatic glucagon receptor (Gs-coupled GPCR), driving glycogenolysis and gluconeogenesis to raise blood glucose.

INDICATIONS

- Severe hypoglycemia (emergency)
- Diagnostic agent for GI motility imaging

DOSING

IM/SC/nasal emergency 1 mg (adult), 0.5 mg (pediatric < 25 kg)

Diagnostic 0.25–2 mg IV

EVIDENCE

Standard-of-care emergency therapy; modern formulations show equivalent efficacy with much improved ease of use.

SAFETY

Nausea, vomiting after administration · Rebound hyperglycemia · Contraindicated in pheochromocytoma, insulinoma

ADVERSE REACTIONS

Nausea and vomiting after administration (very common), hyperglycemia (expected effect), transient hypokalemia, headache. Rare: hypersensitivity, transient hypertension in patients with pheochromocytoma.

Clinical caveat. Newer formulations (Baqsimi nasal, Gvoke ready-to-use pen) eliminated reconstitution at the bedside, simplifying out-of-hospital hypoglycemia rescue.

REGULATORY STATUS FDA-approved · all major regulators

REFERENCES

- Pieber TR et al. Baqsimi PK/PD. Diabetes Care 2015 (intranasal) [not indexed]
Christiansen et al. BMJ Open Diabetes Res Care 2021;9: (PMID 34620618)

Insulin

Humulin · Novolin · Lantus · Levemir · Tresiba · Humalog · NovoLog · Apidra

HISTORY

Isolated by Banting and Best in 1921 at the University of Toronto, the first peptide drug.

SEQUENCE A-chain: GIVEQCCTSICSLYQLENYCN · B-chain:
FVNQHLCGSHLVEALYLVCGERGFFYPKT (disulfide-linked)

51 residues (2 chains)

MECHANISM

Binds the insulin receptor (a tyrosine kinase), triggering GLUT4 translocation to the cell membrane and driving glucose uptake; suppresses hepatic gluconeogenesis.

INDICATIONS

- Type 1 diabetes
- Type 2 diabetes (when oral therapy insufficient)
- Diabetic ketoacidosis

DOSING

Type 1 DM total daily	0.5–1.0 U/kg/day, basal-bolus split
Long-acting basal	Once or twice daily
Rapid-acting prandial	Per meal, weight- and carb-counted

EVIDENCE

Foundational therapy with 100+ years of use. Modern analogues have extensive RCT evidence for glycemic control and outcome benefit.

SAFETY

Hypoglycemia is the principal acute risk · Weight gain · Lipohypertrophy at injection sites

ADVERSE REACTIONS

Hypoglycemia (most common, dose-dependent) · weight gain · injection-site lipohypertrophy or lipoatrophy · local hypersensitivity at injection sites · rare systemic allergy. Severe hypoglycemia is the principal serious AE.

Clinical caveat. Modern analogues (glargine, degludec, lispro, aspart) have replaced NPH and regular insulin for most uses — choice is by pharmacokinetic profile, not generic class.

REGULATORY STATUS FDA-approved · all major regulators

REFERENCES

- Banting FG, Best CH. J Lab Clin Med 1922;7:251 (original isolation) [not indexed]
 Nathan et al. N Engl J Med 1993;329:977–86 (PMID 8366922)
 UKPDS 33. Lancet 1998;352:837 (PMID 9742976) — glycemic control in T2DM

Liraglutide

Victoza · Saxenda

HISTORY

Novo Nordisk's first marketed GLP-1 analogue; FDA-approved for type 2 diabetes (Victoza, 2010) and obesity (Saxenda, 2014).

SEQUENCE HAEGTFTSDVSSYLEGQAAK-[γE-C16]-EFIAWLVGRGRG

31 residues + fatty-acid side chain

MECHANISM

Daily-injection GLP-1 receptor agonist with ~13 hour half-life via albumin binding. Same mechanism as semaglutide; shorter duration.

INDICATIONS

- Type 2 diabetes
- Chronic weight management in adults
- Pediatric obesity (ages 12+)

DOSING

Daily SC (T2DM) 0.6 mg → 1.2 → 1.8 mg

Daily SC (obesity) 0.6 mg → titrate weekly to 3.0 mg

EVIDENCE · GRADE A RATES TYPE 2 DIABETES AND WEIGHT MANAGEMENT

LEADER trial showed MACE reduction in T2DM with CV disease.

SAFETY

GI side effects (typically dose-dependent) · Pancreatitis risk · MTC/MEN2 contraindication

ADVERSE REACTIONS

Most common: nausea (39%), diarrhea (21%), constipation (19%), vomiting (16%), headache, fatigue.

Clinical caveat. Largely displaced by once-weekly agents in adults; no longer the only GLP-1 labeled for pediatric obesity — Wegovy injection also carries a 12-and-over obesity indication; Saxenda's pediatric indication is narrower, restricted to patients over 60 kg. Hypothesis-generating signal for AUD from preclinical and observational data; smaller effect than newer GLP-1 RAs but well-characterized in animal models — under active investigation.

REGULATORY STATUS FDA-approved (Victoza 2010, Saxenda 2014) · EMA

REFERENCES

- Pi-Sunyer X et al. SCALE Obesity. NEJM 2015;373:11 (PMID 26132939)
 Marso SP et al. LEADER. NEJM 2016;375:311 (PMID 27295427)
 Davies MJ et al. SCALE Diabetes. JAMA 2015;314:687 (PMID 26284720)

GRADE A

SMALL-MOLECULE (NON-PEPTIDE) ORAL GLP-1 RA

Orforglipron

Foundayo (Eli Lilly · LY3502970)

HISTORY

Discovered by Chugai, licensed by Lilly in 2018 — the first small-molecule (non-peptide) GLP-1 RA approved.

SEQUENCE Small molecule (LY3502970) — not a peptide

Small molecule (~M.W. 567 Da)

MECHANISM

Once-daily oral non-peptide GLP-1 receptor agonist that binds and activates the human GLP-1 receptor; no food or water restriction (unlike oral semaglutide).

INDICATIONS

- Chronic weight management (obesity / overweight + comorbidity)
- Investigational: type 2 diabetes (ACHIEVE program), OSA, knee OA, hypertension

DOSING

Approved (obesity)	0.8 mg daily → 2.5 → 5.5 mg, ≥30 days per step
Further escalation	9, 14.5, or 17.2 mg daily; max 17.2 mg
T2D	Investigational; trial doses are not an approved regimen

EVIDENCE · GRADE A RATES CHRONIC WEIGHT MANAGEMENT

Multiple positive Phase 3 readouts. ATTAIN-1 (NEJM 2025): 12.4% weight loss at...

SAFETY

GI: nausea, diarrhea, vomiting consistent with injectable GLP-1 class · No new safety signals vs injectable GLP-1 RAs in Phase 3

ADVERSE REACTIONS

Most common: nausea, diarrhea, vomiting, constipation, decreased appetite — consistent with injectable GLP-1 class.

Clinical caveat. First non-peptide GLP-1 RA approved (Foundayo). Major practical advantage over Rybelsus: no food/water restrictions. Beat oral semaglutide head-to-head in ACHIEVE-3. T2D indication pending.

REGULATORY STATUS FDA-approved 1 April 2026 (Foundayo, NDA 220934, Lilly) · obesity or overweight with ≥1 weight-related comorbidity, with reduced-calorie diet and increased activity · T2D not approved

REFERENCES

- Wharton et al. N Engl J Med 2025;393:1796-1806 (PMID 40960239)
 Horn et al. Lancet 2025;406:2927-2944 (PMID 41275875)
 Rosenstock et al. N Engl J Med 2025;393:1065-1076 (PMID 40544435)

Retatrutide

Investigational (Eli Lilly)

HISTORY

Eli Lilly's triple-receptor agonist. Phase 2 (Jastreboff 2023) showed 24%...

SEQUENCE Modified 39-residue peptide with fatty-acid moiety (full sequence proprietary)

39 residues + fatty-acid side chain

MECHANISM

Co-activates GIP, GLP-1, and glucagon receptors. Glucagon-receptor agonism increases hepatic energy expenditure and may...

INDICATIONS

- Obesity (investigational)
- Type 2 diabetes (investigational)
- MASH (investigational)

DOSING

Phase 3 obesity 1 mg → titrate to 12 mg SC weekly

Note Investigational; dosing not finalized

Doses as studied in the trials or protocols named; not an approved label for general use.

EVIDENCE · GRADE A RATES OBESITY, INVESTIGATIONAL

Phase 2 (Jastreboff 2023, n=338): 24.2% weight loss at 48 wks on 12 mg. MASLD substudy (Sanyal 2024): 86% liver-fat reduction.

SAFETY

Same incretin GI profile (nausea, vomiting) · Glucagon component raises modest concern for hepatic glucose output; no clinically apparent issue in Phase 2

ADVERSE REACTIONS

From Phase 2/3: nausea (most common, dose-dependent), vomiting, diarrhea, constipation, decreased appetite.

Clinical caveat. The glucagon component adds hepatic energy expenditure to incretin appetite suppression; whether that beats tirzepatide needs head-to-head data that does not exist.

REGULATORY STATUS *Investigational · TRANSCEND-T2D-1 published; TRIUMPH-1 and -4 unpublished · filing signalled 2026/2027*

REFERENCES

Jastreboff AM et al. Phase 2 obesity. NEJM 2023;389:514 (PMID 37366315)

Rosenstock et al. Lancet 2023;402:529-544 (PMID 37385280)

Sanyal et al. Nat Med 2024;30:2037-2048 (PMID 38858523)

Semaglutide

Ozempic · Wegovy · Rybelsus

HISTORY

Developed by Novo Nordisk; approved by FDA for type 2 diabetes (Ozempic) in 2017, for obesity (Wegovy) in 2021, and as oral therapy (Rybelsus) in 2019.

SEQUENCE H-[Aib]-EGTFTSDVSSYLEGQAAK(γE-C18 diacid)EFIAWLVRGRG-NH₂

31 residues + fatty-acid side chain

MECHANISM

Long-acting GLP-1 receptor agonist; albumin-bound fatty-acid chain extends half-life to ~7 days. Slows gastric emptying, augments glucose-dependent insulin release...

INDICATIONS

- Type 2 diabetes
- Chronic weight management (BMI ≥ 30, or ≥ 27 with comorbidities); obesity from age 12 — Wegovy
- Noncirrhotic MASH with F2–F3 fibrosis, accelerated approval — Wegovy

DOSING

SC weekly (T2DM/obesity) 0.25 mg start; titrate to 2.0 mg (T2DM) or 2.4 mg (obesity)
Oral (Rybelsus) 3 mg → 7 mg → 14 mg PO daily, 30-d titration; fasted, no food/meds 30 min after

EVIDENCE · GRADE A RATES TYPE 2 DIABETES AND WEIGHT MANAGEMENT

STEP 1 (n=1961): 14.9% mean weight loss vs 2.4% placebo at 68 weeks (Wilding 2021).

SAFETY

Nausea, vomiting (titration-dependent) · Pancreatitis risk (rare) · Contraindicated in personal/family history of MTC or MEN2

ADVERSE REACTIONS

Most common: nausea (44%), diarrhea (32%), vomiting (24%), constipation (24%), abdominal pain (20%).

Clinical caveat. Titrate slowly (4-wk increments) to limit nausea-driven discontinuation. Delayed gastric emptying alters absorption of narrow-window oral drugs. AUD signal under investigation (SEMALCO).

REGULATORY STATUS FDA-approved · EMA · most major regulators. Oral tablet formulations carry adult indications only...

REFERENCES

Wilding JPH et al. STEP 1. NEJM 2021;384:989 (PMID 33567185)
 Lincoff AM et al. SELECT. NEJM 2023;389:2221 (PMID 37952131)
 Marso SP et al. SUSTAIN-6. NEJM 2016;375:1834 (PMID 27633186)

Tirzepatide

Mounjaro · Zepbound

HISTORY

Eli Lilly's first dual incretin agonist; FDA-approved for type 2 diabetes (Mounjaro, 2022) and chronic weight management (Zepbound, 2023).

SEQUENCE Y-[Aib]-EGTFTSDYSIYLDKIAQKAFVQWLIAGGPSSGAPPPS · [γE-C20 diacid]
39 residues + fatty-acid side chain

MECHANISM

Activates both GIP and GLP-1 receptors. The dual agonism produces greater appetite suppression and improved insulin sensitivity than GLP-1 monotherapy.

INDICATIONS

- Type 2 diabetes
- Chronic weight management (BMI ≥ 30 or ≥ 27 with comorbidity)
- Obstructive sleep apnea (Zepbound, 2024)

DOSING

SC weekly	2.5 mg → titrate by 2.5 mg every 4 wks
Max dose	15 mg weekly
OSA dose	10–15 mg weekly

EVIDENCE · GRADE A RATES TYPE 2 DIABETES AND WEIGHT MANAGEMENT

SURMOUNT-1 (Jastreboff 2022): 22.5% mean weight loss at 15 mg over 72 weeks.
SURPASS-2: HbA1c reduction of 2.46% (tirzepatide 15 mg) vs 1.86% (semaglutide 1 mg).
Superior weight-loss and glycemic efficacy across the program.

SAFETY

GI: nausea, vomiting, diarrhea · Pancreatitis risk · MTC/MEN2 contraindication

ADVERSE REACTIONS

Most common: nausea (29%), diarrhea (23%), decreased appetite (11%), vomiting (10%), constipation, dyspepsia, abdominal pain.

Clinical caveat. Direct head-to-head Phase 3 data (SURPASS-2) show greater mean weight loss and A1C reduction vs semaglutide 1 mg; whether dual GIP/GLP-1 engagement, dose, or pharmacokinetics drive the difference is not fully resolved.

REGULATORY STATUS FDA-approved · EMA · most major regulators

REFERENCES

- Jastreboff AM et al. SURMOUNT-1. NEJM 2022;387:205 (PMID 35658024)
Frias JP et al. SURPASS-2. NEJM 2021;385:503 (PMID 34170647)
Malhotra et al. N Engl J Med 2024;391:1193-1205 (PMID 38912654)

Pramlintide

Symlin

HISTORY

Synthetic amylin analogue approved by FDA in 2005 for use alongside mealtime insulin. Limited adoption due to need for separate injections.

SEQUENCE KCNTATCATQRLANFLVHSSNNFGPILPPTNVGSNTY-NH₂ (proline substitutions for solubility)

37 residues

MECHANISM

Activates the amylin/CTR-RAMP receptor complex. Slows gastric emptying, suppresses glucagon, and reduces postprandial glucose excursions.

INDICATIONS

- Type 1 diabetes (adjunct to insulin)
- Type 2 diabetes (with insulin)

DOSING

Premeal SC (T1DM) 15 µg → 60 µg per meal

Premeal SC (T2DM) 60 µg → 120 µg per meal

EVIDENCE

Multiple RCTs show modest A1c reduction (~0.4%) and modest weight loss as adjunct to insulin.

SAFETY

Severe hypoglycemia risk when combined with insulin (mandate insulin dose reduction) · Nausea common · Boxed warning for severe insulin-induced hypoglycemia

ADVERSE REACTIONS

Severe insulin-induced hypoglycemia within 3 hours of co-administration (boxed warning). Nausea (28–48%), anorexia (10–17%), vomiting, headache, fatigue, abdominal pain. Mandates insulin dose reduction.

Clinical caveat. Cannot be mixed with insulin in the same syringe — requires a separate injection. Insulin dose must be reduced by 50% at initiation to prevent severe hypoglycemia.

REGULATORY STATUS FDA-approved · Symlin (AstraZeneca) · 2005

REFERENCES

- Whitehouse F et al. Diabetes Care 2002;25:724 (PMID 11919132)
Hollander PA et al. Diabetes Care 2003;26:784 (PMID 12610038)

PART

Growth-Hormone Axis

GHRH analogues, secretagogues, and fragments. Approval status varies sharply within a class that markets itself as one thing.

IN THIS PART

- Somatropin (hGH)
- Tesamorelin
- Sermorelin
- AOD-9604
- CJC-1295
- GHRP-2
- Ipamorelin
- GHRP-6
- Hexarelin
- IGF-1 LR3

GRADE A

RECOMBINANT 191-AMINO-ACID GROWTH HORMONE

Somatropin (hGH)

Genotropin · Norditropin · Humatrope · Saizen · Omnitrope (+ weekly forms)

SEQUENCE 191-residue recombinant human GH

191 residues

MECHANISM

Binds the dimeric GH receptor (JAK2/STAT5), driving hepatic and peripheral IGF-1 production plus direct lipolytic, anabolic, and insulin-antagonizing effects.

INDICATIONS

- Adult growth-hormone deficiency
- Pediatric approved growth indications

DOSING

SC daily (adult GHD)	0.2-0.4 mg/day, titrate to IGF-1; long-acting weekly forms also approved
Pediatric (approved)	0.16-0.7 mg/kg/week divided daily, indication-dependent
Off-label / gray-market	2-6 IU/day SC (about 0.7-2 mg); supraphysiologic, not FDA-validated

EVIDENCE · GRADE A RATES APPROVED GH-DEFICIENCY INDICATIONS

Decades of RCTs establish efficacy for approved GH-deficiency and growth indications.

SAFETY

Edema, arthralgia, myalgia, carpal tunnel · Insulin resistance, hyperglycemia; intracranial hypertension · Contraindicated in active malignancy and acute critical illness

ADVERSE REACTIONS

Most common: fluid retention, arthralgia, myalgia, carpal-tunnel symptoms, glucose intolerance. Supraphysiologic abuse causes acromegaly-like changes, diabetes...

Clinical caveat. Approved replacement for documented GH deficiency, dosed to IGF-1 — not a validated anti-aging or performance drug; supraphysiologic off-label/bodybuilding use is unapproved and WADA-prohibited. The TRIIM epigenetic-age result (Fahy 2019) came from a tiny...

REGULATORY STATUS FDA-approved for GH deficiency and approved growth indications

REFERENCES

- Fahy GM, Brooke RT, Watson JP, et al. Reversal of epigenetic aging and immunosenescent trends in humans. *Aging Cell* 2019;18:e13028 (PMID 31496122)
- Genotropin (somatropin) prescribing information, section 1 — approved GH-deficiency indications
- Yuen KCJ et al. AACE/ACE guidelines for growth hormone use. *Endocr Pract* 2019;25:1191-1232 (PMID 31760824)

Tesamorelin

Egrifta SV · Egrifta WR

HISTORY

A GHRH(1-44) analogue, FDA-approved 2010 as Egrifta for HIV-associated lipodystrophy — the only approved GHRH analogue for a body-composition indication.

SEQUENCE trans-3-hexenoyl-YADAIFTNSYRKVLGQLSARKLLQDIMSRQQGESNQERGARA
RL

44 residues

MECHANISM

GHRH receptor agonist on pituitary somatotrophs; stimulates pulsatile endogenous growth hormone release. Lipolysis in visceral fat appears to be a downstream effect of restored GH/IGF-1 axis.

INDICATIONS

— HIV-associated lipodystrophy (visceral adiposity reduction)

DOSING

SC daily 1.4 mg SC daily (Egrifta SV, 2 mg/vial) or 1.28 mg SC daily (Egrifta WR, 11.6 mg/vial)

EVIDENCE

Two Phase 3 trials pooled in Falutz 2010 (n=806): visceral fat fell 15.4% against placebo at 26 wks, triglycerides also reduced, subcutaneous fat unchanged.

SAFETY

Injection site reactions common · Fluid retention · Glucose intolerance ·
Contraindicated in active malignancy

ADVERSE REACTIONS

Most common: injection-site reactions (erythema, pruritus, pain, irritation, swelling — 25%). Arthralgia, peripheral edema, myalgia, paresthesia.

Clinical caveat. Approved only for HIV-associated lipodystrophy; use for general visceral fat or anti-aging is off-label. It raises IGF-1 and requires monitoring, and benefit reverses on discontinuation — so it is an indefinite commitment, not a course.

REGULATORY STATUS FDA-approved (Egrifta, HIV lipodystrophy, 2010) · EU application withdrawn 2012

REFERENCES

Falutz J et al. J Clin Endocrinol Metab 2010;95:4291-304 (PMID 20554713) — pooled analysis of both Phase 3 trials, n=806, VAT -15.4% vs placebo
Falutz J et al. NEJM 2007;357:2359-70 (PMID 18057338) — first Phase 3, HIV lipodystrophy
Falutz J et al. J Acquir Immune Defic Syndr 2010;53:311-22 (PMID 20101189) — second Phase 3, randomized placebo-controlled

Sermorelin

Geref (discontinued) · compounded

HISTORY

Truncated GHRH analogue, the shortest fragment retaining full biological activity.

SEQUENCE YADAIFTNSYRKVLGQLSARKLLQDIMSR

29 residues

MECHANISM

GHRH receptor agonist; identical mechanism to tesamorelin but with shorter half-life (~10 minutes). Promotes pulsatile endogenous GH release.

INDICATIONS

- Historical: pediatric GH deficiency
- Off-label adult use: marketed for age-related GH/IGF-1 decline (no on-label adult indication)

DOSING

Pediatric (historical)	30 µg/kg SC at bedtime
Adult off-label	100–300 µg SC at bedtime, 5–7 nights/week
Compounded blends	Often paired with ipamorelin or GHRP-6

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE · GRADE B RATES PEDIATRIC GH DEFICIENCY, HISTORICAL

Pediatric efficacy well-established in pre-1990 trials. Adult off-label use lacks RCT support; small open-label studies suggest modest IGF-1 increase.

SAFETY

Injection-site reactions · Headache, flushing · Risk of glucose intolerance with long-term use

ADVERSE REACTIONS

Injection-site reactions (erythema, pain — most common), headache (rare), flushing, dizziness, urticaria, hyperactivity, sleep disturbance.

Clinical caveat. Commonly paired with ipamorelin in compounded blends for additive GHRH (sermorelin) + GHSR (ipamorelin) stimulation. The original brand "Geref" was discontinued in 2008 and the molecule is now compounded-only.

REGULATORY STATUS *Compounded · brand discontinued 2008 · not on not categorized on FDA's 14 May 2026 roster restricted list*

REFERENCES

- Walker RF et al. Pediatrics 1991 — Geref pediatric GHD [not indexed]
 Aleppo G et al. (review) Endocr Pract 1997 — adult GHD diagnostic use [not indexed]

AOD-9604

Research only

HISTORY

Synthetic 16-amino-acid fragment of human growth hormone, developed by Metabolic Pharmaceuticals (Australia) as an anti-obesity drug.

SEQUENCE LRIVQCRSVEGSCGF (with disulfide bridge)

16 residues

MECHANISM

Proposed to bind a putative GH fragment receptor and promote lipolysis without GH receptor activation; mechanism remains poorly characterized.

INDICATIONS

- None approved
- Marketed as nutraceutical in Australia (TGA listed)
- Off-label fat-loss claims

DOSING

Off-label SC 0.5 mg SC twice daily (1 mg/day total)

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Phase 2b obesity trial failed to show significant weight loss vs placebo.

SAFETY

Limited safety data · TGA-listed as a complementary medicine in Australia, not as a drug
· Not approved as a drug anywhere

ADVERSE REACTIONS

Generally well tolerated in Phase 2b obesity trial — adverse-event rates similar to placebo.

Clinical caveat. NO positive RCTs — Phase 2b obesity trial failed its primary endpoint; current use rests on preclinical mechanism work and anecdotal weight-loss claims. Listed as a complementary medicine in Australia rather than approved as...

REGULATORY STATUS *Listed complementary medicine (Australia) · research elsewhere*

REFERENCES

- Stier H et al. (Metabolic Pharmaceuticals) Phase 2b obesity trial — negative primary endpoint [not indexed]
Heffernan MA et al. Endocrinology 2001;142:5182 (preclinical mechanism)

CJC-1295

Research only · compounded

HISTORY

Developed by ConjuChem in the early 2000s; DAC (drug affinity complex) variant uses a maleimidopropionic acid linker to bind albumin in vivo, extending half-life from minutes to ~8 days. Never approved for human use after Phase 2 trials raised safety concerns.

SEQUENCE YADAIFTNSYRKVLAQLSARKLLQDIMSR (CJC-1295 without DAC, "Mod GRF 1-29"); with DAC adds a maleimide-lysine residue

29 residues ($\pm$ DAC linker)

MECHANISM

GHRH receptor agonist with four amino-acid substitutions that resist DPP-4 cleavage.

INDICATIONS

- None approved
- Research and gray-market off-label use: marketed for age-related GH/IGF-1 decline

DOSING

Compounded with DAC 1–2 mg SC weekly (typical anti-aging clinic use)

Without DAC ("Mod GRF") 100 μ g SC once or twice daily

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Phase 1/2 showed sustained IGF-1 elevation; clinical development slowed after safety concerns emerged in related long-acting GHRH-analogue programs (including a fatal event in a CJC-1131 trial that was not definitively attributed to drug).

SAFETY

Sustained high IGF-1 raises theoretical concern for malignancy promotion ·
Injection-site reactions

ADVERSE REACTIONS

Limited human data. Reported: injection-site reactions, flushing, transient headache, mild water retention, numbness or tingling.

Clinical caveat. Evidence base: Phase 1/2 safety pharmacology + animal models + extensive anecdotal off-label use — no positive human efficacy RCTs published. Two distinct forms: 'Mod GRF'...

REGULATORY STATUS Research only · gray market

REFERENCES

- Teichman et al. J Clin Endocrinol Metab 2005;91:799–805 (PMID 16352683)
Ionescu et al. J Clin Endocrinol Metab 2006;91:4792–7 (PMID 17018654)

GHRP-2

Pralmorelin · GHRP Kaken (Japan, diagnostic)

SEQUENCE D-Ala-D-2-Nal-Ala-Trp-D-Phe-Lys-NH₂

6 residues

MECHANISM

GHS-R1a (ghrelin receptor) agonist. More potent GH release than GHRP-6 with somewhat less appetite stimulation, but raises cortisol and prolactin more than ipamorelin.

INDICATIONS

— GH-stimulation testing (Japan, diagnostic)

DOSING

Diagnostic (Japan) 100 ug IV bolus, single dose

Off-label SC 100-300 ug SC 1-3x/day, empty stomach; not FDA-validated

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

GH-stimulation pharmacology well-characterized in humans; pralmorelin is validated as a Japanese GHD diagnostic. No Phase 3 therapeutic trials or Western approval; off-label optimization use rests on anecdote.

SAFETY

Appetite increase (less than GHRP-6) · Cortisol and prolactin elevation · Water retention · FDA 503A Category 3 · 503B Category 2

ADVERSE REACTIONS

Increased appetite (GHSR agonism, milder than GHRP-6), transient cortisol and prolactin rise, water retention, injection-site reactions. Long-term therapeutic safety uncharacterized.

Clinical caveat. The only GHRP with a regulatory approval — pralmorelin is a licensed GH-stimulation diagnostic in Japan, not a therapy. Off-label optimization use is unstudied and raises cortisol/prolactin more than ipamorelin, which is preferred for clean GH release. 503A Category 3 (503B 503A Category 3 (503B Category 2)) (compounding-restricted).

REGULATORY STATUS *Approved in Japan as a GH-stimulation diagnostic; research or gray-market elsewhere*

REFERENCES

Bowers CY. Growth hormone-releasing peptide (GHRP). Cell Mol Life Sci 1998;54:1316-29 (PMID 9893708)

Ipamorelin

Research only · compounded

HISTORY

Developed by Helsinn Healthcare in the late 1990s as a selective growth hormone secretagogue. Phase 2 trials for postoperative ileus terminated 2011; never approved.

SEQUENCE Aib-His-D-2-Nal-D-Phe-Lys-NH₂

5 residues (modified)

MECHANISM

Selective ghrelin/GH secretagogue receptor (GHSR-1a) agonist. Unlike GHRP-6, does not stimulate cortisol, ACTH, or prolactin meaningfully.

INDICATIONS

- None approved
- Off-label / gray-market: marketed for age-related GH/IGF-1 decline and athletic use

DOSING

Off-label SC 100–200 µg SC, 5 of 7 days (anti-aging clinic protocol)

Combined with CJC-1295 For synergistic GHRH/GHSR stimulation

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Phase 2 showed effective GH release but failed primary endpoint (postoperative ileus). No long-term safety or efficacy data in adult anti-aging populations.

SAFETY

Generally well-tolerated short-term · Long-term effects unknown · GHSR agonism increases appetite · 503A nomination withdrawn · 503B Category 2

ADVERSE REACTIONS

Generally well tolerated short term in research and off-label use. Reported: mild headache, transient flushing, increased appetite (GHSR agonism)...

Clinical caveat. Evidence base: Phase 2 (postoperative ileus) failed its primary endpoint; otherwise Phase 1 safety data + decades of anecdotal off-label GH-secretagogue use. Almost always co-administered with a GHRH analogue (CJC-1295 or sermorelin) — alone, GH release is modest. In combination, the dual GHRH + GHSR signal is synergistic.

REGULATORY STATUS Research only · gray market

REFERENCES

- Raun et al. Eur J Endocrinol 1998;139:552–61 (PMID 9849822)
Beck et al. Int J Colorectal Dis 2014;29:1527–34 (PMID 25331030)

GHRP-6

Research only · compounded

SEQUENCE His-D-Trp-Ala-Trp-D-Phe-Lys-NH₂

6 residues

MECHANISM

GHS-R1a (ghrelin receptor) agonist. Potent GH release, but associated with marked appetite stimulation, and it raises cortisol, prolactin, and ACTH more than ipamorelin.

INDICATIONS

— No approved indication

DOSING

Off-label SC 100–300 ug SC 1–3x/day, empty stomach; reported use only

Stacking Often paired with a GHRH analogue (Mod GRF / CJC-1295)

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Historically pivotal in characterizing the ghrelin/GHS-R axis; demonstrated GH release in human studies. No Phase 3 efficacy or approval; off-label use rests on anecdote and the marked appetite effect.

SAFETY

Marked appetite stimulation · Cortisol, prolactin, and ACTH elevation · Water retention · FDA 503A Category 3 · 503B Category 2

ADVERSE REACTIONS

Pronounced hunger (strongest of the GHRPs), transient cortisol/prolactin/ACTH rise, water retention, injection-site reactions, tingling. Long-term effects uncharacterized.

Clinical caveat. Mainly of historical and research interest; its strong appetite stimulation and HPA/prolactin activation make it less suitable than ipamorelin for GH optimization, though it is sometimes used deliberately as an appetite stimulant. Usually paired with a GHRH analogue. 503A Category 3 (503B 503A Category 3 (503B Category 2)) (compounding-restricted).

REGULATORY STATUS No approved indication; research or compounded use only

REFERENCES

Bowers CY, Momany F, Reynolds GA, Hong A. On the in vitro and in vivo activity of a new synthetic hexapeptide that acts on the pituitary to specifically release growth hormone. *Endocrinology* 1984;114:1537–45 (PMID 6714155)

Hexarelin

Research only

HISTORY

Developed by Mediolanum Farmaceutici in the 1990s; reached Phase 2 for cardiac and pediatric indications. Never approved. Distinctive for its potent GH release but...

SEQUENCE H-D-MeTyr-D-Trp-Ala-Trp-D-Phe-Lys-NH₂

6 residues (modified)

MECHANISM

GHSR-1a agonist, more potent than ipamorelin or GHRP-6 but less selective; also activates CD36 on cardiomyocytes.

INDICATIONS

- None approved
- Investigational cardiac applications

DOSING

Research/off-label 100–200 µg SC, 1–3 times daily

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Small clinical studies suggest possible cardioprotective effects via CD36, but no Phase 3 evidence. Tachyphylaxis develops quickly with repeated dosing.

SAFETY

Cortisol and prolactin elevation · Tachyphylaxis · No long-term data · Not approved anywhere

ADVERSE REACTIONS

Increased ACTH and cortisol (HPA-axis activation), prolactin elevation, tachyphylaxis with repeated dosing, fatigue and sleep changes. More HPA effect than ipamorelin makes hexarelin less suitable for repeated use.

Clinical caveat. Tachyphylaxis with repeated dosing limits practical utility. Ipamorelin produces a cleaner pharmacologic profile (no ACTH or cortisol elevation) for most off-label GH-axis purposes. Trial stage: Phase 2 GH-stimulation testing (1990s); commercially abandoned.

REGULATORY STATUS Research only · gray-market use · no human therapeutic approval

REFERENCES

Imbimbo BP et al. JCEM 1994;78:693 — first GH stimulation studies

Ghigo E et al. JCEM 1994;79:1407 — Italian GHRP-6 family pharmacology [not indexed]

GRADE D

IGF-1 RECEPTOR AGONIST · SYNTHETIC GROWTH-FACTOR ANALOGUE

IGF-1 LR3

Research only · gray-market (various manufacturers)

SEQUENCE Long R3 IGF-1 (83-residue analogue)

83 residues

MECHANISM

Binds IGF-1R (a receptor tyrosine kinase), driving PI3K/AKT/mTOR and MAPK/ERK signaling — muscle protein synthesis, satellite-cell activation, glucose uptake, nitrogen retention, proliferation, and fat oxidation — independently of growth hormone.

INDICATIONS

— No approved indication

DOSING

Gray-market SC/IM

20-100 ug post-workout or at target muscle; reported use only

Cycle

4-6 weeks; doses extrapolated from community practice, not validated

No published human source exists for these figures: no approved product, no trial record, no human study. Recorded as reported gray-market practice only.

EVIDENCE

No human RCTs for body-composition or performance. Animal models show IGF-1R-mediated muscle hypertrophy, satellite-cell activation...

SAFETY

Hypoglycemia — chief acute risk (insulin-receptor cross-reactivity) · Acromegalic-pattern soft-tissue enlargement with prolonged use · Theoretical oncogenic risk (IGF-1R overexpressed in cancers)

ADVERSE REACTIONS

Gray-market use: hypoglycemia (carry fast-acting glucose, monitor), joint/jaw pain, facial puffiness, water retention. Prolonged use may cause acromegaly-like soft-tissue enlargement. Theoretical tumor-promotion risk.

Clinical caveat. GRADE D — animal/preclinical and in-vitro only; no human efficacy trials, and healthy-aging claims are speculative. Gray-market purity is unverified and counterfeiting common (some products contain insulin). Chief acute risk: hypoglycemia from insulin-receptor cross-reactivity; long-term IGF-1R stimulation carries an unquantifiable oncogenic risk. Not licensed for human use.

REGULATORY STATUS No approved indication; gray-market only

REFERENCES

Dominikowski A et al. Peptides marketed for performance and longevity: absence of human studies. 2025 (PMID 42395176)

PART

Regenerative & Repair

Repair peptides, almost all research-only. This is where preclinical evidence and clinical marketing diverge most.

IN THIS PART

- ARA-290
- Thymosin α 1
- BPC-157
- Follistatin
- KPV
- TB-500 (Ac-LKKTETQ)
- Thymulin

ARA-290

Cibinetide · Araim Pharmaceuticals (investigational)

HISTORY

Designed by Anthony Cerami and Michael Brines at Araim Pharmaceuticals. ARA-290 is an 11-amino-acid fragment of erythropoietin's B-helix domain — the region that binds the innate-repair receptor...

SEQUENCE QEQLERALNSS

11 residues

MECHANISM

Activates the innate-repair receptor (IRR), a heterodimer of EPOR and CD131 (βCR), expressed on damaged tissue (neurons, endothelium, cardiomyocytes) but not on erythroid progenitors.

INDICATIONS

- None approved
- Investigational: sarcoid small-fiber neuropathy (Phase 2 positive)
- Investigational: diabetic peripheral neuropathy, neuropathic pain syndromes

DOSING

Investigational SC 4 mg SC four times weekly (per Phase 2 protocols)

Doses as studied in the trials or protocols named; not an approved label for general use.

EVIDENCE · GRADE B RATES SARCOID SMALL-FIBRE NEUROPATHY

Phase 2 in sarcoidosis small-fiber neuropathy (Dahan et al., Mol Med 2013) showed significant reduction in spontaneous pain vs placebo.

SAFETY

- Well-tolerated in Phase 2 · No erythropoietic effects observed (hemoglobin unchanged)
- Long-term safety still pending Phase 3 data

ADVERSE REACTIONS

Generally well tolerated. Mild injection-site reactions and headache reported. No erythropoietic effects detected; no thrombotic events in trial corpus to date.

Clinical caveat. ARA-290 is meaningfully distinct from the gray-market peptides on this list — it has a coherent receptor pharmacology, IND-cleared trials, and a defined commercial sponsor (Araim). Not yet FDA-approved.

REGULATORY STATUS *Investigational (Phase 2) · Araim Pharmaceuticals · not approved*

REFERENCES

- Heij L et al. Mol Med 2012;18:1430 (PMID 23168581) — first chronic pain Phase 1
 Dahan A et al. Mol Med 2013;19:334 — sarcoid small-fiber neuropathy Phase 2

GRADE B

THYMIC PEPTIDE (IMMUNOMODULATOR)

Thymosin α 1

Zadaxin · Thymalfasin

HISTORY

Isolated from thymic extracts by Goldstein et al. in 1977.

SEQUENCE Ac-SDAAVDTSSIEITTKDLKEKKEVVEEAEN

28 residues (N-acetylated)

MECHANISM

Modulates T-cell and dendritic-cell function; enhances Th1 responses via TLR9 signaling on plasmacytoid dendritic cells.

INDICATIONS

- Hepatitis B/C (in approved countries)
- Immunomodulation adjunct to vaccines
- Cancer adjunct in some protocols

DOSING

SC twice weekly

1.6 mg per dose, typically 6+ months

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE · GRADE B RATES HEPATITIS B/C WHERE APPROVED

Multiple RCTs for chronic hepatitis B/C show modest virologic response improvement.

SAFETY

Well-tolerated; injection site reactions most common · Autoimmune disease theoretical concern · Pregnancy category not established

ADVERSE REACTIONS

Generally well tolerated. Most common: injection-site reactions (transient), mild fatigue, chills. Transient elevation in autoantibody titers reported.

Clinical caveat. Holds FDA orphan drug designations (hepatocellular carcinoma, melanoma stage IIb–IV, DiGeorge anomaly) but no NDA marketing approval — these are different regulatory categories that are commonly conflated.

REGULATORY STATUS

Marketed in 35+ countries · FDA orphan drug designations only · removed from nomination withdrawn; in no 503A category on FDA's 14 May 2026 roster bulk substances list in Sept 2024 (PCAC review pending)

REFERENCES

Camps E et al. — hepatitis B trials (multiple, foreign-approved) [not indexed]

Wu J et al. Crit Care 2013;17:R8 — sepsis trial

Garaci E et al. Ann N Y Acad Sci 2007;1112:14 — pharmacology review

BPC-157

Bepecin · BPC 157 Stable · compounded

HISTORY

Synthetic pentadecapeptide derived from a sequence within a human gastric juice protein, first characterized by the Zagreb group (Sikiric et al.) in 1993.

SEQUENCE GEPPPGKPADDAGLV

15 residues

MECHANISM

Pleiotropic. Reported actions include VEGFR2 upregulation, eNOS activation, FAK-paxillin signaling modulation, GH receptor upregulation in fibroblasts and tendon.

INDICATIONS

- None approved
- Off-label/anecdotal: tendon and ligament injuries, IBD, ulcers, post-operative recovery

DOSING

Off-label SC/IM 250–500 µg once or twice daily

Off-label oral 500 µg once or twice daily

Topical cream (burns) 1–2% cream/gel BID; superficial burns only

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Extensive rodent literature, primarily from one research group (Sikiric, Zagreb).

SAFETY

Short-term tolerability good in limited human use; long-term safety not characterized · WADA Section S0 (all routes, no TUE)

ADVERSE REACTIONS

Limited human data. Reported by users and limited trials: injection-site reactions (mild), temporary fatigue, headache...

Clinical caveat. No high-quality human efficacy trials — all clinical use rests on preclinical (rodent) data and anecdote. Four routes: SC injection...

REGULATORY STATUS *nomination withdrawn; in no 503A category on FDA's 14 May 2026 roster · not approved · July 2026 PCAC review (UC)*

REFERENCES

Sikirić et al. J Physiol Paris 1993;87:313-27 (PMID 8298609)

Sikiric et al. Curr Pharm Des 2018;24:1990-2001 (PMID 29879879)

Lee et al. Altern Ther Health Med 2021;27:8-13 (PMID 34324435)

Follistatin

Research only (recombinant) · investigational gene therapy

HISTORY

Endogenous glycoprotein first isolated in 1987 from ovarian follicular fluid (hence the name).

SEQUENCE Full glycoprotein — 315 or 344 residues depending on isoform
315–344 residues (glycosylated)

MECHANISM

Binds and neutralizes activin A and myostatin (GDF-8). Myostatin is the principal negative regulator of muscle growth...

INDICATIONS

- None approved
- Investigational: Duchenne muscular dystrophy (AAV-mediated gene therapy)
- Off-label gray-market: muscle hypertrophy, athletic enhancement

DOSING

Gray-market recombinant 100–200 µg SC every other day for 10–30 day cycles

AAV-FST gene therapy One-time intramuscular delivery (investigational)

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Strong animal data for muscle hypertrophy with viral-vector delivery.

SAFETY

Theoretical cardiac hypertrophy concerns (myostatin loss in animal models) ·
Tendon–muscle imbalance · Long-term immunogenicity unknown

ADVERSE REACTIONS

Limited human data outside DMD gene-therapy trials. Reported in gray-market use:
injection-site reactions, mild flushing.

Clinical caveat. Evidence base: investigational gene-therapy Phase 1/2 for Duchenne; the injectable peptide form has no published human efficacy data — anecdotal use only. Two delivery routes are very different — AAV-mediated gene therapy...

REGULATORY STATUS Research only · gray market (recombinant) · Phase 1/2 for AAV-FST gene therapy in DMD

REFERENCES

- Mendell et al. Mol Ther 2017;25:870–879 (PMID 28279643)
Kota J et al. Sci Transl Med 2009;1:6ra15 — preclinical primate

KPV

Research only · gray market

HISTORY

Lysine-Proline-Valine, the C-terminal tripeptide of α -MSH (residues 11–13).

SEQUENCE KPV

3 residues

MECHANISM

Anti-inflammatory; modulates NF- κ B signaling and inflammatory cytokine production without binding MC receptors. Specific molecular target remains uncharacterized.

INDICATIONS

- None approved
- Off-label/gray-market: inflammatory conditions, IBD, skin inflammation, post-procedure healing

DOSING

Off-label SC 200–500 μ g daily

Off-label oral 1–5 mg daily (absorption limited)

Common stack Combined with BPC-157 for tissue-healing protocols

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Preclinical data for inflammatory-bowel-disease models; small open-label studies for atopic dermatitis. No randomized human trials.

SAFETY

Generally well-tolerated · Injection-site reactions possible · Long-term safety not characterized

ADVERSE REACTIONS

Limited human data. Generally well tolerated; injection-site reactions reported. No serious AEs documented in animal or limited human use.

Clinical caveat. Evidence base: animal IBD models + small dermatology case reports + anecdotal use (often within the KLOW stack) — no human RCTs published. Frequently sold as the 'KLOW stack'...

REGULATORY STATUS *Research only · gray market · July 2026 PCAC review (wound healing, inflammatory conditions)*

REFERENCES

- Kannengiesser K et al. Inflamm Bowel Dis 2008;14:324 — DSS-colitis model
 Dalmaso et al. Gastroenterology 2007;134:166–78 (PMID 18061177)

TB-500 (Ac-LKKTETQ)

Research only · compounded

HISTORY

A synthetic heptapeptide, Ac-LKKTETQ, corresponding to residues 17–23 of mature thymosin β 4 (UniProt P62328) — its central actin-binding domain.

SEQUENCE Ac-LKKTETQ — residues 17–23 of thymosin β 4 (parent: SDKPDMAEIEKFDKSLKKTETQEKNP LPSKETIEQEQAGES)

7 residues (parent thymosin β 4 is 43)

MECHANISM

Sequesters monomeric G-actin, modulating cytoskeletal dynamics. Promotes endothelial cell migration, angiogenesis...

INDICATIONS

- None approved
- Off-label: soft tissue injury, wound healing

DOSING

Off-label SC	2–10 mg per week (1–2 doses)
Loading phase	5 mg twice weekly for 4 weeks, then maintenance
Stack with BPC-157	250 μ g SC daily or 5 days/week (commonly co-administered)

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Phase 2 trials for ophthalmic indications missed primary endpoints.

SAFETY

Generally well-tolerated · Long-term safety unknown

ADVERSE REACTIONS

Limited safety data in humans. Reported by users: injection-site reactions, transient fatigue, head rush, lightheadedness, lethargy.

Clinical caveat. TB-500 and thymosin β 4 are not interchangeable: the heptapeptide is a fragment, and vendor products labelled TB500 or TB1000 have been found inconsistent with their own labelling. Ask what a patient actually bought.

REGULATORY STATUS Research only · gray market · WADA-prohibited · July 2026 PCAC review (wound healing)

REFERENCES

Crockford D, Turjman N, Allan C, Angel J. Thymosin β 4: structure, function, and biological properties. *Ann N Y Acad Sci* 2010;1194 (PMID 20536467)

Thymulin

Facteur Thymique Sérique (FTS) · research compound

HISTORY

Identified as Facteur Thymique Sérique (FTS) by Bach and Dardenne in 1977 (Nature 1977;266:55).

SEQUENCE pGlu-Ala-Lys-Ser-Gln-Gly-Gly-Ser-Asn (Zn²⁺ binding required for activity)
9 residues

MECHANISM

Active (zinc-bound) thymulin binds receptors on T-lymphocytes and NK cells, promoting T-cell differentiation and modulating cytokine balance (IL-2, IFN- γ).

INDICATIONS

- None approved
- Investigational: age-related immune decline, post-chemo immune reconstitution; off-label gray-market: marketed for longevity claims

DOSING

Off-label SC 50–100 μ g twice weekly (informal anti-aging protocols)
Note Active form requires zinc — co-administration or adequate zinc status assumed

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Decades of preclinical work characterizing the zinc-thymulin axis (Bach, Dardenne, Goldstein groups).

SAFETY

No modern human safety data · Generally well-tolerated in historical clinical use

ADVERSE REACTIONS

Reported in historical clinical use: well tolerated, occasional injection-site reactions. No major safety signals at physiologic-range...

Clinical caveat. Evidence base: extensive preclinical characterization (Bach/Dardenne, 1977–2000s) + small historical clinical studies in primary immunodeficiency and aging — no modern RCTs, no current clinical use anywhere.

REGULATORY STATUS Research only · gray market · unapproved

REFERENCES

- Bach et al. Nature 1977;266:55–7 (PMID 300146)
Dardenne M et al. PNAS 1982;79:5370 — zinc-dependent activation of thymulin
Reggiani PC et al. Curr Pharm Des 2014;20:4690 — thymulin in immunity and aging (review)

PART

Dermatology & Cosmetic

Topical and injected cosmetic peptides. Route matters to FDA's safety assessment here, though it does not change drug status.

IN THIS PART

- GHK
- GHK-Cu
- Argireline
- Leuphasyl
- Matrixyl 3000
- PTD-DBM
- Palmitoyl Tripeptide-1
- Zinc Thymulin

GHK

Research compound · cosmetic ingredient

HISTORY

Glycyl-L-histidyl-L-lysine, identified by Pickart and Thaler in 1973 in human plasma.

SEQUENCE GHK

3 residues (uncomplexed)

MECHANISM

Same fundamental biology as GHK-Cu — binds endogenous copper from albumin or extracellular pools to deliver copper to enzymes like superoxide dismutase and lysyl oxidase. Activates wound-healing and antioxidant gene-expression programs.

INDICATIONS

- None approved
- Cosmetic: skin appearance, hair
- Off-label/investigational: oncology adjunct claims (limited evidence)

DOSING

Off-label SC 2.5 mg SC three times weekly (gray-market oncology protocol)

Cosmetic topical 0.1–2% in serum or cream formulations

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Same in-vitro literature as GHK-Cu (Pickart et al., 2015, 2017).

SAFETY

Generally well tolerated topically · Injection-site reactions similar to but milder than GHK-Cu · No copper-overload concern (no pre-complexed copper)

ADVERSE REACTIONS

Mild injection-site irritation; occasional rash. Reportedly better-tolerated SC than GHK-Cu because the copper is not pre-loaded onto the peptide.

Clinical caveat. Evidence base: limited direct human data; mechanism is inferred from the GHK-Cu literature and topical cosmetic studies — injectable use rests on anecdote. Sometimes marketed as having distinct activity from GHK-Cu — in practice the two interconvert in tissue depending on local copper availability.

REGULATORY STATUS *Research only · not regulated as a drug*

REFERENCES

- Pickart et al. Nat New Biol 1973;243:85–7 (PMID 4349963)
 Pickart et al. Biomed Res Int 2015;2015:648108 (PMID 26236730)

GHK-Cu

Cosmetic ingredient · research compound

HISTORY

Glycyl-L-histidyl-L-lysine, identified in human plasma in 1973 by Pickart and Thaler. Its copper-binding form (GHK-Cu) has been studied...

SEQUENCE GHK

3 residues (copper-complexed)

MECHANISM

Copper-binding tripeptide; modulates copper delivery to enzymes including superoxide dismutase and lysyl oxidase. Activates wound-healing gene expression programs via copper-dependent signaling.

INDICATIONS

- Cosmetic: skin appearance, hair
- No drug approvals

DOSING

Topical cosmetic	0.05–2% formulations
Off-label injectable	1–2 mg SC, 5–7 days per week

EVIDENCE

Decades of dermatologic research (Pickart et al., 2015, 2017) support roles in collagen synthesis, antioxidant defense, and skin/hair benefits.

SAFETY

Well-tolerated topically · Injection-site pain and skin rash common with SC use · Hypersensitivity reactions reported but uncommon

ADVERSE REACTIONS

Topical: mild erythema, dryness, or irritation at higher concentrations; contact dermatitis in sensitized users. Injectable: injection-site pain...

Clinical caveat. Evidence base: topical cosmetic RCTs (cosmeceutical use) and preclinical wound-healing models — injectable human use rests on anecdote, no RCTs. The only peptide in this guide with a recognized US compounding pathway...

REGULATORY STATUS FDA 503A Category 1 (non-injectable, US) · NHPID approved (Canada) · Category 1 for non-injectable routes; injectable nomination withdrawn (injectable)

REFERENCES

- Pickart et al. *Nat New Biol* 1973;243:85–7 (PMID 4349963)
 Pickart L, Margolina A. *Int J Mol Sci* 2018;19:1987 (PMID 29986520) — comprehensive review
 Leyden J et al. (multiple cosmetic RCTs, 2002 onward, manufacturer-sponsored) [not indexed]

GRADE D

SNAP-25 N-TERMINAL MIMETIC ("TOPICAL BOTULINUM")

Argireline

Acetyl Hexapeptide-3 · Acetyl Hexapeptide-8 (INCI)

HISTORY

Developed by Lipotec (Barcelona) in the late 1990s. Designed to mimic the N-terminus of SNAP-25 — the SNARE protein cleaved by botulinum toxin...

SEQUENCE Ac-EEMQRR-NH₂

6 residues (N-acetylated, C-amidated)

MECHANISM

Mimics the N-terminus of SNAP-25, competing with native SNAP-25 for incorporation into the SNARE complex. In theory, reduces vesicle fusion at the neuromuscular junction, attenuating muscle contraction and expression lines.

INDICATIONS

- None approved (cosmetic ingredient only)
- Cosmetic: expression lines, fine wrinkles (forehead, peri-orbital)

DOSING

Cosmetic topical

5–10% concentration in serum or cream, twice daily

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Multiple manufacturer-sponsored cosmetic studies showing modest improvement in wrinkle depth. Independent evidence less conclusive — transdermal penetration to facial muscle is implausible at topical concentrations...

SAFETY

Cosmetic ingredient with two decades of broad use · Generally well-tolerated topically · Minimal systemic exposure

ADVERSE REACTIONS

Rare contact dermatitis. No serious adverse events in cosmetic use. Theoretical concern about systemic SNARE inhibition is not borne out...

Clinical caveat. Marketed as "topical botox" but the mechanism requires the peptide to reach the neuromuscular junction beneath facial muscles — most dermatology authorities consider this implausible at cosmetic concentrations. Cosmetic effect, if real, is modest and not a substitute for botulinum injections.

REGULATORY STATUS Cosmetic ingredient · not a drug

REFERENCES

Blanes-Mira C et al. Int J Cosmet Sci 2002;24:303 — original cosmetic SNAP-25 mimic
Wang Y et al. (manufacturer follow-ups, multiple) [not indexed]

GRADE D

STABILIZED ENKEPHALIN PENTAPEPTIDE (COSMETIC)

Leuphasyl

Pentapeptide-18 (INCI) · enkephalin analog

HISTORY

Developed by Lipotec as a cosmetic enkephalin analog. Premise: enkephalin signaling at the neuromuscular junction modulates acetylcholine release, so a topical enkephalin analog should reduce facial muscle contraction synergistically with Argireline.

SEQUENCE Tyr-D-Ala-Gly-Phe-Leu-NH₂

5 residues (D-Ala stabilized)

MECHANISM

Binds enkephalin receptors (presumed δ -opioid receptors) at the neuromuscular junction, attenuating acetylcholine release and thereby reducing muscle contraction. Same caveat as Argireline regarding topical penetration to the neuromuscular junction.

INDICATIONS

- None approved (cosmetic ingredient only)
- Cosmetic: expression lines, fine wrinkles (typically combined with Argireline)

DOSING

Cosmetic topical

0.5% in serum or cream, twice daily; often combined with Argireline for additive effect

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Manufacturer-sponsored cosmetic studies suggest modest synergy with Argireline. Independent evidence is limited. Frequently sold as a finished combination product rather than a stand-alone ingredient.

SAFETY

Cosmetic ingredient · Generally well-tolerated topically · Minimal systemic exposure

ADVERSE REACTIONS

Rare contact dermatitis. No serious adverse events reported in cosmetic use.

Clinical caveat. Like Argireline, the topical-botox premise depends on transdermal delivery to the neuromuscular junction, which is mechanistically implausible at cosmetic concentrations. Cosmetic effect is modest at best and synergistic claims with Argireline rest on company-sponsored data. Trial stage: manufacturer-sponsored cosmetic studies only.

REGULATORY STATUS Cosmetic ingredient · not a drug

REFERENCES

Lipotec (manufacturer) cosmetic studies — limited independent peer-reviewed evidence [not indexed]

Matrixyl 3000

Matrixyl 3000 (Sederma) · cosmetic

SEQUENCE pal-GHK + pal-GQPR blend
tripeptide + tetrapeptide blend

MECHANISM

Matrikines act as ECM-fragment signals: pal-GHK and pal-GQPR stimulate fibroblast synthesis of collagen (I, III), fibronectin, hyaluronic acid, and glycosaminoglycans, and damp IL-6-type inflammation. Palmitoylation aids penetration. Topical.

INDICATIONS

— Topical wrinkle reduction, skin firmness (cosmetic)

DOSING

Topical 2-8% of the trade-name blend in serum or cream,
1-2x/day

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Manufacturer and small cosmetic studies report reduced wrinkle depth and volume over 1-2 months vs placebo; independent peer-reviewed RCTs are limited. No drug-level efficacy data.

SAFETY

Topical use generally well tolerated · Mild irritation or contact dermatitis possible · Cosmetic ingredient — no drug oversight

ADVERSE REACTIONS

Topical use is generally well tolerated; occasional mild irritation, redness, or contact sensitivity. No systemic exposure expected from cosmetic concentrations.

Clinical caveat. A cosmetic matrikine blend (pal-GHK + pal-GQPR), not a drug. Topical only; effect size is modest and formulation-dependent.

REGULATORY STATUS Cosmetic ingredient, not a drug; topical only

REFERENCES

Sederma (Croda) technical dossier, Matrixyl 3000 (pal-GHK + pal-GQPR) [not indexed]

PTD-DBM

Topical hair-growth research compound

HISTORY

Developed by the Choi group at Yonsei University (published 2017). Fuses the HIV-Tat protein transduction domain (PTD) to...

SEQUENCE YGRKKRRQRRR-PEDVPDYGFAGCPP

~24 residues (TAT-PTD + DBM)

MECHANISM

Penetrates hair-follicle cells via the TAT-PTD domain. Inside the cell, the DBM peptide disrupts CXXC5-Dishevelled binding...

INDICATIONS

- None approved
- Off-label/compounded: androgenetic alopecia, post-chemotherapy alopecia (topical)

DOSING

Off-label topical

0.1–0.5% concentration in spray vehicle, applied to scalp 1–2× daily; often combined with valproic acid 5% for synergistic Wnt activation

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Preclinical and limited dermatology research from the Yonsei group (Choi et al., 2017). No large RCT.

SAFETY

Topical route limits systemic exposure · Scalp irritation possible · Long-term effects unknown

ADVERSE REACTIONS

Scalp irritation, mild redness, itching. Rarely allergic contact dermatitis. Hair-shedding phase during early use as anagen induction begins (similar to minoxidil).

Clinical caveat. Mechanistically the most interesting investigational hair-loss compound on this list, but human efficacy data is thin — most use is via compounding pharmacies. Effective only combined with a Wnt-pathway modulator (typically valproic acid). Trial stage: preclinical only; no human RCTs.

REGULATORY STATUS *Research only · used in compounding pharmacies for hair growth*

REFERENCES

Lee et al. J Invest Dermatol 2017;137:2260–2269 (PMID 28595998)

Palmitoyl Tripeptide-1

Pal-GHK · cosmetic ingredient

SEQUENCE Pal-Gly-His-Lys

3 residues

MECHANISM

Matrikine that signals dermal fibroblasts to upregulate collagen (types I, III, IV), fibronectin, and glycosaminoglycans, mimicking ECM-fragment repair signaling. Palmitoylation enhances skin penetration. Topical.

INDICATIONS

— Topical anti-wrinkle, skin firmness, photoaging (cosmetic)

DOSING

Topical

Trace concentrations (about 1-10 ppm) in serums or creams, 1-2x/day

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Cosmetic in-vitro and small clinical studies report increased collagen synthesis and modest wrinkle improvement; no independent drug-level RCTs.

SAFETY

Topical use generally well tolerated · Rare mild irritation or sensitivity · Cosmetic ingredient — no drug oversight

ADVERSE REACTIONS

Generally well tolerated topically; rare mild irritation or contact sensitivity. No systemic effects expected at cosmetic concentrations.

Clinical caveat. The palmitoylated GHK matrikine at the core of Matrixyl-type blends; cosmetic, not a drug. Used at very low (ppm) concentrations; efficacy is modest and formulation-dependent, supported by cosmetic studies rather than independent RCTs. Topical only — no injectable use.

REGULATORY STATUS *Cosmetic ingredient, not a drug; topical only*

REFERENCES

Lintner K, Peschard O. Biologically active peptides: from a laboratory bench curiosity to a functional skin care product. *Int J Cosmet Sci* 2000;22:207-18 (PMID 18503476)

Zinc Thymulin

Compounded · research compound

HISTORY

Thymulin is a nonapeptide thymic hormone discovered by Bach and Dardenne in the 1970s, requiring zinc binding for activity. Originally investigated for immunomodulation.

SEQUENCE pGlu-Ala-Lys-Ser-Gln-Gly-Gly-Ser-Asn

9 residues (zinc-coordinated)

MECHANISM

Parenterally, modulates T-cell maturation and thymopoiesis. Topical hair-growth mechanism less defined — proposed to optimize follicular microenvironment and dermal-papilla signaling.

INDICATIONS

- None approved
- Off-label topical: androgenetic alopecia (typically combined with GHK-Cu)

DOSING

Off-label topical 0.2% in 30 mL topiclick or spray vehicle, 1 mL applied to scalp once daily

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Preclinical and small case-series for hair growth. No published RCT. Most clinical use is in compounded combination products with GHK-Cu and other adjuncts...

SAFETY

Topical route limits systemic exposure · Scalp irritation occasional · No long-term safety data

ADVERSE REACTIONS

Mild scalp irritation, occasional hyperpigmentation. No serious adverse events documented in case-series use.

Clinical caveat. Reasonable adjunct in combination topical hair products where each component is well-tolerated; not a stand-alone therapy with proven efficacy. The role of the zinc cofactor itself complicates the question of whether 'zinc thymulin' acts as a peptide or as a vehicle for topical zinc.

REGULATORY STATUS *Research only · compounded for cosmetic hair use*

REFERENCES

Bach et al. Nature 1977;266:55-7 (PMID 300146)

Caroleo MC et al. (zinc thymulin hair-loss case series, small) [not indexed]

PART

Bone Metabolism

Anabolic and antiresorptive peptides with approved indications.

IN THIS PART

- Abaloparatide
- Calcitonin (salmon)
- Teriparatide

Abaloparatide

Tymlos

HISTORY

Synthetic analogue of parathyroid hormone-related protein (PTHrP) 1-34; FDA-approved 2017 for postmenopausal osteoporosis. Developed by Radius Health.

SEQUENCE AVSEHQLLHDKGKSIQDLRRRELLEKLLAKLHTA-NH2 (with modifications)
34 residues

MECHANISM

Selectively activates the RG (rapid, reversible) conformation of PTH1R; theoretically more anabolic than teriparatide with less hypercalcemia risk.

INDICATIONS

- Postmenopausal osteoporosis (high fracture risk)
- Men with osteoporosis at high fracture risk

DOSING

SC daily 80 µg once daily

Max duration 2 years lifetime

EVIDENCE

ACTIVE trial: 86% reduction in new vertebral fracture vs placebo over 18 months. ACTIVEExtend follow-on with alendronate showed durable benefit.

SAFETY

Boxed warning for osteosarcoma (class effect) · Hypercalcemia (less than teriparatide) · Orthostatic hypotension · Same contraindications as teriparatide

ADVERSE REACTIONS

Most common: hypercalciuria (11%), dizziness (10%), nausea (8%), headache (8%), palpitations, fatigue, abdominal pain, vertigo. Less hypercalcemia than teriparatide. Same boxed osteosarcoma class warning.

Clinical caveat. PTHrP analogue with less hypercalcemia than teriparatide but the same 24-month lifetime limit and osteosarcoma class warning. Real-world adoption limited by cost and prior-authorization barriers.

REGULATORY STATUS FDA-approved (Tymlos, 2017) · EMA (Eladynos, 2022)

REFERENCES

Miller et al. JAMA 2016;316:722-33 (PMID 27533157)

Cosman F et al. ACTIVEExtend. Mayo Clin Proc 2017;92:200 (PMID 28160873)

Calcitonin (salmon)

Miacalcin · Fortical

HISTORY

Salmon calcitonin is 30–50× more potent than human calcitonin in humans.
FDA-approved as Miacalcin: injection 1986...

SEQUENCE CSNLSTCVLGKLSQELHKLQTYPRNTNTGSGTP-NH₂ (with intramolecular disulfide)

32 residues

MECHANISM

Inhibits osteoclast activity via the calcitonin receptor (a Gs-coupled GPCR). Also has central analgesic effects via spinal calcitonin receptors.

INDICATIONS

- Paget disease
- Hypercalcemia of malignancy
- Postmenopausal osteoporosis (second-line, intranasal use)
- Vertebral compression fracture pain

DOSING

Intranasal (osteoporosis)	200 IU one nostril daily, alternating
SC/IM (Paget)	100 IU daily, then maintenance
SC/IM (hypercalcemia)	4–8 IU/kg q6–12h

EVIDENCE · GRADE A RATES PAGET DISEASE AND HYPERCALCAEMIA

Modest fracture-reduction effect in PROOF trial (~33% vertebral); no effect on hip or non-vertebral fractures. EMA restricted long-term use in 2012 due to malignancy signal.

SAFETY

Possible increase in malignancy with long-term use (EMA review) · Nasal irritation · GI side effects with parenteral use

ADVERSE REACTIONS

Intranasal: rhinitis, nasal irritation, epistaxis. Parenteral: nausea, vomiting, flushing, injection-site reactions. Increased malignancy signal with chronic use...

Clinical caveat. Intranasal use was restricted by EMA in 2012 due to a malignancy signal in long-term users. Now primarily short-term for hypercalcemia, Paget disease, and vertebral compression fracture pain.

REGULATORY STATUS FDA-approved · EMA (restricted)

REFERENCES

Chesnut CH et al. PROOF. Am J Med 2000;109:267 (PMID 10996576) — fracture prevention
Cochrane review (Wells GA et al. 2008) — modest fracture-prevention efficacy [not indexed]

Teriparatide

Forteo · Bonsity · biosimilars

HISTORY

Recombinant N-terminal 34-amino-acid fragment of parathyroid hormone. FDA-approved 2002 for severe osteoporosis. The first anabolic (rather than antiresorptive) osteoporosis therapy.

SEQUENCE SVSEIQLMHNLGKHLNSMERVEWLRKKLQDVHNF

34 residues

MECHANISM

Intermittent PTH receptor (PTH1R) activation on osteoblasts; pulsed exposure favors bone formation over resorption (opposite of chronically elevated PTH).

INDICATIONS

- Severe osteoporosis (postmenopausal women, men, glucocorticoid-induced)
- Patients at high risk of fracture

DOSING

SC daily 20 µg once daily

Max duration 2 years lifetime

EVIDENCE · GRADE A RATES SEVERE OSTEOPOROSIS

Pivotal Fracture Prevention Trial showed 65% reduction in vertebral fracture and 53% reduction in non-vertebral fracture vs placebo.

SAFETY

Boxed warning for osteosarcoma (rat data; not seen in human postmarketing) · Hypercalcemia, hypercalciuria · Orthostatic hypotension · Contraindicated in prior radiation, Paget disease, bone malignancy

ADVERSE REACTIONS

Most common: nausea (8%), headache, dizziness, leg cramps, arthralgia. Transient hypercalcemia and hypercalciuria. Orthostatic hypotension after first doses. Boxed warning for osteosarcoma (rat data; not clearly seen in human postmarketing).

Clinical caveat. Lifetime 24-month maximum due to rat osteosarcoma signal (not seen in 20+ years of human postmarketing). Sequencing teriparatide → bisphosphonate preserves the anabolic bone gains after discontinuation.

REGULATORY STATUS FDA-approved · EMA · all major regulators

REFERENCES

- Neer et al. N Engl J Med 2001;344:1434-41 (PMID 11346808)
 Saag KG et al. (vs alendronate) NEJM 2007;357:2028 (PMID 18003959)

PART

Somatostatin Analogues

Approved agents for acromegaly and neuroendocrine tumours.

IN THIS PART

- Lanreotide
- Octreotide
- Pasireotide

Lanreotide

Somatuline Depot

HISTORY

Long-acting somatostatin analogue from Ipsen; FDA-approved 2007 for acromegaly and 2014 for neuroendocrine tumors. The CLARINET trial established its role in slowing progression of advanced GEP-NETs.

SEQUENCE Nal-Cys-Tyr-D-Trp-Lys-Val-Cys-Thr-NH₂ (disulfide-bridged)
8 residues (cyclic)

MECHANISM

Same as octreotide; SSTR2 and SSTR5 agonist, with depot formulation providing month-long exposure.

INDICATIONS

- Acromegaly
- Locally advanced or metastatic GEP-NET
- Carcinoid syndrome

DOSING

Deep SC monthly 120 mg every 4 weeks (typical maintenance)

EVIDENCE

CLARINET trial: significantly prolonged progression-free survival in metastatic GEP-NET (HR 0.47). Equivalent efficacy to octreotide LAR for acromegaly.

SAFETY

GI side effects · Gallbladder issues · Glucose dysregulation · Injection-site reactions

ADVERSE REACTIONS

Most common: diarrhea (49%), cholelithiasis (27%), abdominal pain, flatulence, nausea, injection-site reactions. Hyperglycemia, bradycardia. Long-term gallbladder disease is a class effect of somatostatin analogues.

Clinical caveat. Equivalent efficacy to octreotide LAR for acromegaly; the pre-filled deep-SC depot syringe is easier for patients to self-administer than octreotide LAR.

REGULATORY STATUS FDA-approved (Somatuline Depot, 2007) · EMA

REFERENCES

Caplin ME et al. CLARINET. NEJM 2014;371:224 (PMID 25014687) — pivotal NET
Vinik et al. Endocr Pract 2016;22:1068-80 (PMID 27214300)

Octreotide

Sandostatin · Sandostatin LAR · Mycapssa

HISTORY

First somatostatin analogue, developed by Sandoz (now Novartis); FDA-approved 1988. Oral formulation (Mycapssa) approved 2020.

SEQUENCE D-Phe-Cys-Phe-D-Trp-Lys-Thr-Cys-Thr-ol (disulfide-bridged)

8 residues (cyclic)

MECHANISM

Agonist at somatostatin receptors SSTR2 and SSTR5 (less so SSTR3); inhibits hormone secretion (GH, insulin, glucagon, gastrin) and reduces neuroendocrine tumor proliferation.

INDICATIONS

- Acromegaly
- Carcinoid syndrome and VIPoma
- Gastrointestinal bleeding from varices
- Diagnostic (Octreoscan)

DOSING

SC TID (acute) 100–600 µg/day in divided doses

IM monthly (LAR) 10–40 mg every 4 weeks

Oral (Mycapssa) 20 mg twice daily, titrate to 40–80 mg/day

EVIDENCE

Multiple RCTs and decades of clinical experience for acromegaly biochemical control (~50–60%) and carcinoid syndrome symptom relief.

SAFETY

GI side effects (steatorrhea, nausea, gallstones) · Hypothyroidism · Hyperglycemia or hypoglycemia

ADVERSE REACTIONS

Most common: diarrhea (34%), abdominal cramps (28%), nausea (27%), steatorrhea. Cholelithiasis (gallstones) in 10–20% with chronic use.

Clinical caveat. Mycapssa (oral, 2020) is the first oral somatostatin analogue. Most patients still use IM LAR for convenience and durability; the radiolabeled Octreoscan form remains a standard neuroendocrine-tumor imaging agent.

REGULATORY STATUS FDA-approved · EMA · all major regulators

REFERENCES

Rinke A et al. PROMID. JCO 2009;27:4656 (PMID 19704057) — midgut NET
Caplin ME et al. CLARINET. NEJM 2014;371:224 (PMID 25014687)

Pasireotide

Signifor · Signifor LAR

HISTORY

Novartis's broader-spectrum somatostatin analogue, approved 2012 for Cushing disease and 2014 for acromegaly. Binds SSTR1, SSTR2, SSTR3, and SSTR5 (octreotide and lanreotide are SSTR2-selective).

SEQUENCE Cyclo[(4R)-4-(2-aminoethyl-carbamoyloxy)-L-prolyl-L-phenylglycyl-D-tryptophyl-L-lysyl-O-benzyl-L-tyrosyl-L-phenylalanyl]

6 residues (cyclic)

MECHANISM

Pan-SSTR agonist with highest affinity for SSTR5 (relevant to ACTH-secreting pituitary adenomas in Cushing disease).

INDICATIONS

- Cushing disease (when surgery has failed or is contraindicated)
- Acromegaly (octreotide/lanreotide-resistant)

DOSING

SC twice daily (Cushing) 0.3–0.9 mg BID (start 0.6 or 0.9 mg, titrate)

IM monthly (LAR) 20–60 mg every 4 weeks

EVIDENCE

Phase 3 for Cushing: ~25% biochemical normalization. PAOLA trial in acromegaly showed superiority to octreotide/lanreotide in patients with inadequate response.

SAFETY

Hyperglycemia is the principal limiting toxicity (much more common than with octreotide/lanreotide) · GI side effects · QT prolongation · Cholelithiasis

ADVERSE REACTIONS

Most common: hyperglycemia (much more than octreotide/lanreotide — 73% in Cushing patients), diarrhea (58%), cholelithiasis (30%), nausea, hypocortisolism, QT prolongation. Hyperglycemia frequently requires anti-diabetic therapy.

Clinical caveat. Hyperglycemia is the dose-limiting toxicity, occurring in approximately 73% of Cushing patients — much more than octreotide or lanreotide. The pan-SSTR profile (especially SSTR5) explains both efficacy in Cushing and the metabolic adverse-event pattern.

REGULATORY STATUS FDA-approved (Signifor, 2012) · EMA

REFERENCES

- Gadelha et al. *Lancet Diabetes Endocrinol* 2014;2:875–84 (PMID 25260838)
Colao A et al. *NEJM* 2012;366:914 (PMID 22397653) — Cushing disease

PART

Melanocortin & Sexual

Melanocortin agonists, from approved drugs to the gray-market analogue that shares their receptor and none of their oversight.

IN THIS PART

- Afamelanotide
- Bremelanotide
- Melanotan II
- PT-141

Afamelanotide

Scenesse

HISTORY

α -MSH analogue developed by Clinuvel; EMA-approved 2014 and FDA-approved 2019 for erythropoietic protoporphyria (EPP), a rare disorder causing extreme photosensitivity. The first approved melanocortin therapy.

SEQUENCE Ac-Ser-Tyr-Ser-Nle-Glu-His-D-Phe-Arg-Trp-Gly-Lys-Pro-Val-NH₂
13 residues

MECHANISM

Synthetic α -MSH analogue, selectively activates MC1R on melanocytes to drive eumelanin production. Increased skin melanin protects against UV-induced phototoxicity.

INDICATIONS

— Erythropoietic protoporphyria (EPP)

DOSING

Implant 16 mg slow-release implant subcutaneously every 2 months

EVIDENCE

Three pivotal Phase 3 trials (CUV029, CUV030, CUV039) across 244 EPP adults at 22 US/EU sites. Langendonk 2015 (NEJM) showed significantly greater pain-free time in direct sunlight. 10-year post-authorization safety data show no new signals.

SAFETY

Site reactions at implant · Headache, nausea · Skin pigmentation increase (expected effect) · Nevus darkening; dermatologic surveillance recommended

ADVERSE REACTIONS

Most common: implant-site reactions (erythema, pain, hyperpigmentation — 28%), nausea, fatigue, headache, back pain, melanocytic naevus darkening, nasopharyngitis. Dermatologic surveillance recommended.

Clinical caveat. The approval covers only the Clinuvel-manufactured 16 mg bioresorbable implant (Scenesse) administered every 2 months for EPP — not generic afamelanotide powder, compounded forms, or research-grade Melanotan 1 vials. Conflating Scenesse approval with cosmetic-tanning melanotan products is a common and consequential misrepresentation of the regulatory record.

REGULATORY STATUS FDA-approved as Scenesse (NDA 210797, Oct 2019) for EPP only · EMA-approved 2014 (exceptional circumstances)

REFERENCES

Langendonk JG et al. NEJM 2015;373:48 (PMID 26132941) — Scenesse pivotal EPP
Biolcati G et al. Br J Dermatol 2015;172:1601 — open-label long-term EPP data

Bremelanotide

Vyleesi

HISTORY

Developed by Palatin Technologies; FDA-approved 2019 for premenopausal hypoactive sexual desire disorder. Originally developed as a nasal spray (PT-141), reformulated to SC autoinjector after blood pressure concerns.

SEQUENCE Ac-Nle-cyclo(Asp-His-D-Phe-Arg-Trp-Lys)-OH

7 residues (cyclic)

MECHANISM

Melanocortin receptor (MC1R, MC3R, MC4R) agonist. MC4R activation in central nervous system thought to drive sexual desire effect.

INDICATIONS

— Hypoactive sexual desire disorder (HSDD) in premenopausal women

DOSING

SC as-needed

1.75 mg ~45 min before sexual activity

Max

One dose per 24 hours, 8 doses per month

EVIDENCE

RECONNECT trials (n>1200): modest improvement in desire and distress scores vs placebo (~25% responders vs 17%).

SAFETY

Transient BP elevation · Nausea (in 40%) · Focal hyperpigmentation · Contraindicated in uncontrolled hypertension and CV disease

ADVERSE REACTIONS

Most common: nausea (40%), flushing (20%), injection-site reactions (13%), headache (11%), vomiting (5%). Transient blood-pressure elevation (peaks at 2–4h post-dose). Focal hyperpigmentation of face/gums/breasts with repeated use.

Clinical caveat. On-demand SC injection 45+ minutes before sexual activity, with a maximum of 1 dose per 24 hours and 8 per month. Transient BP elevation peaks 2–4 hours post-dose; pre-existing hypertension should be addressed first.

REGULATORY STATUS FDA-approved as Vyleesi (HSDD in premenopausal women, 2019)

REFERENCES

Kingsberg SA et al. RECONNECT. *Obstet Gynecol* 2019;134:899 (PMID 31599840)
Clayton et al. *J Womens Health (Larchmt)* 2022;31:171-182 (PMID 35147466)

Melanotan II

Gray market only

HISTORY

Synthesized at the University of Arizona in the 1980s as a tanning agent.

SEQUENCE Ac-Nle-cyclo(Asp-His-D-Phe-Arg-Trp-Lys)-NH₂

7 residues (cyclic)

MECHANISM

Non-selective MC1R/MC3R/MC4R/MC5R agonist. MC1R stimulation drives melanogenesis (tanning); MC4R stimulation drives sexual side effects.

INDICATIONS

- None approved
- Off-label/gray-market: tanning

DOSING

Loading SC 150 µg SC daily for 1–2 weeks (titrating up)

Maintenance SC 500 µg SC twice weekly after loading

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Human exposure is limited to two small early studies: a single-blind placebo-controlled pilot Phase I in three male volunteers, dose-limiting somnolence at 0.03 mg/kg (Dorr 1996), and a 10-man erectile-dysfunction crossover study (Wessells 1998).

SAFETY

Nausea, flushing common · Spontaneous penile erection / priapism

ADVERSE REACTIONS

Documented in case reports and series: nausea, flushing, spontaneous penile erection or priapism, darkening of pre-existing nevi, new nevus formation, cases of melanoma after use, rhabdomyolysis, encephalopathy. No regulatory oversight of purity or dose.

Clinical caveat. Not approved anywhere and not the same molecule as afamelanotide. The tanning use is unstudied; melanoma, priapism...

REGULATORY STATUS *Not approved · gray market · purchase banned in several countries*

REFERENCES

Dorr RT et al. Life Sci 1996;58:1777 — original University of Arizona Phase I

Multiple case reports of melanoma, priapism, rhabdomyolysis (see TGA/MHRA warnings) [not indexed]

Wessells H et al. J Urol 1998;160:389–93 (PMID 9679884) — 10-man double-blind crossover, psychogenic ED

GRADE C

GRAY-MARKET INTRANASAL USE OF BREMELANOTIDE MOLECULE

PT-141

Research only · gray market intranasal

HISTORY

Palatin Technologies research code for what became bremelanotide.

SEQUENCE Ac-Nle-cyclo(Asp-His-D-Phe-Arg-Trp-Lys)-OH

7 residues (cyclic) — same molecule as bremelanotide

MECHANISM

Same molecule as bremelanotide — melanocortin receptor (MC1R/MC3R/MC4R) agonist. MC4R activation in CNS drives sexual desire effect.

INDICATIONS

- None approved under the PT-141 designation
- Off-label gray-market: male erectile dysfunction, sexual desire in either sex (intranasal)

DOSING

Intranasal (gray-market)

5–20 mg per dose, ~45 min before activity

Note

No validated protocol; original Palatin nasal program discontinued

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

No published trials of intranasal PT-141 at gray-market doses.

SAFETY

Transient BP elevation (peaks 2–4 h post-dose) — concern at original intranasal doses · Nausea, focal hyperpigmentation · Avoid in uncontrolled hypertension

ADVERSE REACTIONS

Adverse profile similar to bremelanotide: nausea, flushing, headache, spontaneous erection/priapism in men, focal hyperpigmentation with repeated use.

Clinical caveat. Evidence base: anecdotal gray-market intranasal use only — no published trials at current doses. Original Palatin nasal program halted at Phase 2 for BP concerns. For clinical use, default to bremelanotide (Vyleesi) SC.

REGULATORY STATUS *Research only · gray market intranasal (SC form = Bremelanotide)*

REFERENCES

Diamond et al. J Sex Med 2006;3:628–638 (PMID 16839319)

Safarinejad et al. J Urol 2008;179:1066–71 (PMID 18206919)

See bremelanotide for the FDA-approved subcutaneous formulation data [not indexed]

PART

Reproductive & Pituitary

Gonadotropins, GnRH agonists, and posterior pituitary peptides.

IN THIS PART

- Desmopressin
- Goserelin
- Leuprolide
- Oxytocin
- hCG
- Kisspeptin-10

Desmopressin

DDAVP · Stimate · Nocdurna

HISTORY

Synthetic analogue of vasopressin developed in the 1960s, FDA-approved 1978 for diabetes insipidus. Selectively activates V2 (antidiuretic) receptors without...

SEQUENCE 1-deamino-Cys-Tyr-Phe-Gln-Asn-Cys-Pro-D-Arg-Gly-NH₂ (with disulfide)
9 residues

MECHANISM

Selective V2 receptor agonist on renal collecting duct cells; promotes aquaporin-2 insertion and water reabsorption. Also stimulates von Willebrand factor release from endothelial Weibel-Palade bodies.

INDICATIONS

- Central diabetes insipidus
- Nocturnal enuresis
- Mild hemophilia A and von Willebrand disease (boost vWF)
- Nocturia in adults

DOSING

Intranasal (DI) 5–40 µg/day in 1–2 divided doses

Oral (DI/enuresis) 0.1–0.4 mg/day

IV (vWD/hemophilia A) 0.3 µg/kg over 30 min

EVIDENCE

Standard-of-care therapy for central DI for >40 years. Modest efficacy for nocturnal enuresis.

SAFETY

Hyponatremia is the principal serious risk · Headache · Avoid in patients with type IIB vWD (paradoxical thrombocytopenia)

ADVERSE REACTIONS

Most serious: hyponatremia (can be severe, with seizures). Headache, nausea, abdominal pain, facial flushing. Water intoxication if free water intake...

Clinical caveat. Sodium monitoring is essential — hyponatremia can be severe and is the principal serious AE. Type IIB von Willebrand disease is a contraindication (paradoxical thrombocytopenia).

REGULATORY STATUS FDA-approved · all major regulators

REFERENCES

Robson WL et al. J Pediatr 1996;128:769 — nocturnal enuresis [not indexed]
Bertholini et al. Anaesth Intensive Care 2000;28:199–201 (PMID 10788975)

Goserelin

Zoladex

HISTORY

AstraZeneca's GnRH agonist depot, FDA-approved 1989. Unique cylindrical implant formulation pre-loaded in a syringe for subcutaneous abdominal placement.

SEQUENCE pGlu-His-Trp-Ser-Tyr-D-Ser(t-Bu)-Leu-Arg-Pro-Aza-Gly-NH₂
10 residues (modified)

MECHANISM

Same mechanism as leuprolide; sustained GnRH receptor activation causing pituitary desensitization.

INDICATIONS

- Prostate cancer
- Breast cancer (premenopausal)
- Endometriosis
- Uterine fibroids

DOSING

Implant 3.6 mg every 28 days, or 10.8 mg every 12 weeks

EVIDENCE

Equivalent efficacy to leuprolide for prostate cancer; preferred for breast cancer ovarian suppression based on SOFT/TEXT trials.

SAFETY

Same as leuprolide (hot flashes, BMD loss, sexual dysfunction) · Implant site reactions · Tumor flare with initial dose

ADVERSE REACTIONS

Hot flashes (62%), decreased libido, vaginal dryness (in women), BMD loss, mood changes, weight changes, headache, implant-site reactions. Tumor flare with initial dosing in prostate cancer.

Clinical caveat. Implant administration (vs leuprolide IM/SC) gives reliable depot release without reconstitution. SOFT and TEXT trials established it as the preferred GnRH agonist for premenopausal breast-cancer ovarian suppression.

REGULATORY STATUS FDA-approved (Zoladex, 1989) · EMA · all major regulators

REFERENCES

Goldhirsch A et al. ZIPP. Lancet Oncol 2001 — adjuvant breast cancer [not indexed]
Klijn JG et al. JCO 2001;19:343 — premenopausal MBC

Leuprolide

Lupron · Eligard · Lupron Depot

HISTORY

First clinically used GnRH agonist; FDA-approved 1985 for advanced prostate cancer. Depot formulations enable monthly to 6-month dosing.

SEQUENCE pGlu-His-Trp-Ser-Tyr-D-Leu-Leu-Arg-Pro-NH-Et

9 residues (modified)

MECHANISM

Initial GnRH receptor agonism causes LH/FSH surge ("flare"); sustained exposure desensitizes pituitary gonadotropes, leading to profound suppression of LH, FSH, testosterone, and estradiol.

INDICATIONS

- Advanced prostate cancer
- Endometriosis and uterine fibroids
- Central precocious puberty
- Gender-affirming care (puberty blocker)

DOSING

Daily SC	1 mg daily
Monthly depot IM/SC	7.5 mg every 4 weeks
3-month depot	22.5 mg

EVIDENCE

Standard of care for advanced prostate cancer for decades; survival benefit established in combination with antiandrogens.

SAFETY

Initial testosterone surge can worsen disease (use with antiandrogen) · Hot flashes, sexual dysfunction, BMD loss, metabolic changes · Injection-site reactions

ADVERSE REACTIONS

Most common: hot flashes (55%), decreased libido, erectile dysfunction, BMD loss with chronic use, weight gain, fatigue, mood changes, gynecomastia, injection-site reactions.

Clinical caveat. Initial testosterone surge ("flare") requires antiandrogen coverage for the first 2–4 weeks in prostate cancer to prevent disease flare. Depot and daily formulations are clinically interchangeable.

REGULATORY STATUS FDA-approved · EMA · all major regulators

REFERENCES

Crawford ED et al. NEJM 1989;321:419 — prostate cancer
Lupron Depot trials (multiple) for endometriosis, central precocious puberty [not indexed]

Oxytocin

Pitocin · Syntocinon

HISTORY

Isolated by Vincent du Vigneaud in 1953, the first peptide hormone to be sequenced and synthesized (Nobel Prize 1955).

SEQUENCE CYIQNCPLG-NH₂ (with intramolecular disulfide)

9 residues

MECHANISM

Activates oxytocin receptors on uterine myometrium to drive contraction; also acts on mammary tissue (let-down reflex) and centrally on bonding-related neural circuits.

INDICATIONS

- Production of uterine contractions in the third stage of labor
- Control of postpartum bleeding or hemorrhage

DOSING

IV labor induction	0.5–6 mU/min, titrate to contractions
IV/IM postpartum	10 units IM or 10–40 units IV diluted
Intranasal (off-label)	16–40 IU (typically 24 IU); historical 4 IU/nostril Syntocinon (discontinued 2013)

EVIDENCE · GRADE A RATES LABOUR INDUCTION AND POSTPARTUM HAEMORRHAGE

Cornerstone of labor management for >60 years. Autism, anxiety, and bonding trials have largely failed primary endpoints.

SAFETY

Uterine hyperstimulation, fetal distress · Hypotension with rapid IV bolus

ADVERSE REACTIONS

Labor use: uterine hyperstimulation, fetal distress (most concerning), uterine rupture (rare). Hypotension or arrhythmias with rapid IV...

Clinical caveat. Off-label use for autism, social bonding, and anxiety has largely failed in trials — poor blood-brain-barrier penetration limits CNS effects of both peripheral and intranasal oxytocin. Obstetric uses remain the only evidence-based indications.

REGULATORY STATUS FDA-approved · EMA · all major regulators

REFERENCES

- Carter et al. *Annu Rev Psychol* 2013;65:17–39 (PMID 24050183)
 Yatawara et al. *Mol Psychiatry* 2015;21:1225–31 (PMID 26503762)
 Pitocin (oxytocin injection, USP) prescribing information — indicated to produce uterine contractions during the third stage of labor and to control postpartum bleeding

hCG

Pregnyl · Novarel · Ovidrel (recombinant)

HISTORY

Isolated from the urine of pregnant women in the 1920s.

SEQUENCE Heterodimer: α -subunit (92 aa, shared with LH/FSH/TSH) + β -subunit (145 aa, unique to hCG)

237 residues across 2 glycosylated chains

MECHANISM

Binds the LH/CG receptor (LHCGR) on Leydig cells (in men) and theca/corpus luteum cells (in women).

INDICATIONS

- Cryptorchidism (pediatric)
- Hypogonadotropic hypogonadism (male)
- Ovulation induction (anovulatory infertility)

DOSING

Cryptorchidism	500–4000 IU IM weekly × 4–12 weeks
Male hypogonadism	500–2000 IU IM 2–3× weekly
IVF trigger	5000–10000 IU IM once (or recombinant Ovidrel 250 µg SC once)

EVIDENCE · GRADE A RATES APPROVED ENDOCRINE AND FERTILITY USES

Decades of clinical use; well-established efficacy for ovulation induction and male hypogonadism. Strong evidence for IVF triggering...

SAFETY

Ovarian hyperstimulation syndrome (OHSS) is the principal serious risk ·
Thromboembolic events · Contraindicated in hormone-sensitive cancers

ADVERSE REACTIONS

Most common: injection-site pain, fatigue, headache, mood changes, breast tenderness.
Serious: ovarian hyperstimulation syndrome...

Clinical caveat. Misused historically in the "HCG diet" (Simeons protocol) — repeatedly debunked; weight loss attributable to severe caloric restriction, not hCG. In men, used off-label to preserve testicular function during testosterone replacement therapy and to restore spermatogenesis post-cycle.

REGULATORY STATUS FDA-approved · all major regulators

REFERENCES

- Trimble EL et al. (older fertility-treatment literature) [not indexed]
Brinholi FF et al. Front Endocrinol 2017 — induction-of-ovulation reviews [not indexed]

Kisspeptin-10

Investigational

HISTORY

Originally identified in 1996 as the product of the KISS1 gene — initially named 'metastin' for its metastasis-suppressor activity.

SEQUENCE YNWNSFGLRF-NH₂

10 residues (active fragment of kisspeptin-54)

MECHANISM

Activates the KISS1R (GPR54) receptor on GnRH neurons in the hypothalamus, triggering pulsatile GnRH release.

INDICATIONS

- Investigational: hypogonadotropic hypogonadism
- Investigational: IVF oocyte maturation trigger (alternative to hCG to reduce OHSS risk)
- Investigational: constitutional delayed puberty

DOSING

Diagnostic GnRH testing 1–10 µg/kg IV bolus or SC

IVF trigger (Phase 3) 12.8 mg SC once

Doses as studied in the trials or protocols named; not an approved label for general use.

EVIDENCE · GRADE B RATES INVESTIGATIONAL REPRODUCTIVE ENDOCRINE USES

Phase 2/3 data for IVF triggering shows efficacy with reduced ovarian hyperstimulation syndrome (OHSS) risk vs hCG.

SAFETY

Generally well-tolerated · Mild flushing and headache reported · Long-term safety not yet evaluated

ADVERSE REACTIONS

Most common: mild flushing (most patients), transient headache, injection-site reactions. Generally well tolerated; no major safety signals in completed Phase 3 trials.

Clinical caveat. Acts UPSTREAM of GnRH (at the hypothalamus, not the pituitary) — restoring physiologic GnRH pulsatility rather than overriding it like leuprolide. Most promising near-term clinical application is replacing hCG as an IVF trigger to...

REGULATORY STATUS *Investigational (Phase 3 for IVF triggering) · not approved*

REFERENCES

Dhillon WS et al. JCEM 2005;90:6609 (PMID 16174713) — first human studies
Skorupskaitė et al. Hum Reprod Update 2014;20:485–500 (PMID 24615662)

PART

Mitochondrial & Longevity

Mitochondrial-derived and geroprotective peptides. Human evidence thins rapidly through this part.

IN THIS PART

- SS-31 (Elamipretide)
- Epitalon
- Klotho
- MOTS-c
- Vilon
- 5-Amino-1MQ
- FOXO4-DRI
- G-Klotho Frag 33
- Humanin

SS-31 (Elamipretide)

Forzinity (Stealth BioTherapeutics)

HISTORY

Synthetic mitochondria-targeted tetrapeptide developed by Hazel Szeto. FDA accelerated approval Sept 19...

SEQUENCE D-Arg-Tyr(2,6-diMe)-Lys-Phe-NH₂

4 residues (modified)

MECHANISM

Accumulates in mitochondrial inner membrane via cardiolipin binding; preserves mitochondrial cristae structure, reduces ROS production, and maintains ATP synthesis under stress.

INDICATIONS

- Barth syndrome (FDA-approved, ≥30 kg) — to improve muscle strength
- Investigational: primary mitochondrial myopathy, geographic atrophy, heart failure

DOSING

Barth (FDA-approved) 40 mg SC once daily (≥30 kg)

Renal (eGFR <30) 20 mg SC daily (not on dialysis)

EVIDENCE · GRADE B RATES BARTH SYNDROME

TAZPOWER (N=12, randomized crossover + 192-week open-label extension): primary endpoints (6-minute walk, fatigue score) NOT met.

SAFETY

Injection-site reactions universal (100% in trial) · Hypersensitivity reactions reported — skin and respiratory

ADVERSE REACTIONS

Injection-site reactions universal (100%): erythema, pain, induration, pruritus, bruising, urticaria. Hypersensitivity — skin...

Clinical caveat. First FDA-approved mitochondria-targeted therapeutic. Approval was granted on an intermediate endpoint (knee extensor strength) in a 12-patient trial where both primary endpoints (6MWT, fatigue score) were not met; confirmatory trial required. Expansion beyond Barth syndrome remains uncertain.

REGULATORY STATUS FDA-approved (accelerated, Sept 19 2025) · Forzinity · Stealth BioTherapeutics

REFERENCES

Thompson et al. Genet Med 2024;26:101138 (PMID 38602181)

Karaa et al. Neurology 2023;101:e238-e252 (PMID 37268435)

FDA label (NDA 215244, 2025) — Forzinity prescribing information [not indexed]

Epitalon

Research only

HISTORY

Epitalon, Epithalon and AEDG are the same synthetic tetrapeptide under different spellings and trade names.

SEQUENCE AEDG (Ala-Glu-Asp-Gly) — synthetic tetrapeptide

4 residues

MECHANISM

Proposed to upregulate telomerase activity and extend telomere length in vitro.

INDICATIONS

- None Western-approved
- Russia: gerontology / anti-aging applications

DOSING

Off-label SC 5–10 mg daily for 10–20 day courses, 2–3 times per year (typical Russian protocols)

Off-label oral 50–100 mg daily for 10–20 day courses (Khavinson capsule protocol)

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Studies almost entirely from Khavinson group; not independently replicated.

SAFETY

Limited Western safety data · Reportedly well-tolerated in Russian use

ADVERSE REACTIONS

Reported in Russian clinical and off-label use: generally well tolerated, occasional mild fatigue, sleep changes, transient injection-site reactions.

Clinical caveat. Naming matters at the point of purchase: Epitalon, Epithalon and AEDG name one synthetic tetrapeptide, while Epithalamin, Endoluten and Pineamin are animal-derived pineal extracts, not the peptide and not endogenous human secretions. Evidence rests almost entirely on one Russian group's preclinical and small clinical work, with no Western RCTs.

REGULATORY STATUS Research only outside Russia · July 2026 PCAC review (insomnia)

REFERENCES

Anisimov et al. Biogerontology 2003;4:193-202 (PMID 14501183)

Khavinson VKh, Morozov VG. multiple Russian publications, 1980s–2010s [not indexed]

GRADE C

ANTI-AGING GLYCOPROTEIN (RECOMBINANT FRAGMENTS)

Klotho

Research compound · investigational

HISTORY

Named for the Greek fate who spins the thread of life; klotho-deficient mice were described in 1997 with a syndrome resembling accelerated ageing.

SEQUENCE Proprietary fragments (20–40 residues); full α -Klotho is 1012 residues
Variable – full protein 1012 residues; injectable fragments 20–40 residues

MECHANISM

Membrane klotho is an obligate co-receptor for FGF23 in phosphate handling; a cleaved soluble form circulates and is proposed to modulate insulin/IGF-1 and Wnt signalling.

INDICATIONS

- None established
- Research use only: ageing, cognition, and CKD biology

DOSING

Off-label SC 0.5–2 mg twice weekly (highly variable, no validated protocol)

Note Marketed fragments vary in length and purity; clinical effects unestablished

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE · GRADE C RATES ENDOGENOUS KLOTHO BIOLOGY, NOT MARKETED FRAGMENTS

Strong observational and preclinical data — klotho-deficient mice show accelerated ageing, and human KL-VS variants associate with cognition and longevity.

SAFETY

No human safety data for exogenous Klotho administration · Theoretical concern: altered phosphate handling and FGF23 signaling · Long-term effects unknown

ADVERSE REACTIONS

Not characterized in humans. Animal studies have not shown major safety signals.

Clinical caveat. No exogenous klotho product exists in an approved or investigational form for humans. “Klotho” sold as a peptide is not the 130 kDa protein the literature studies, so the evidence base does not transfer to it.

REGULATORY STATUS *Research only · gray market · unapproved*

REFERENCES

Kuro-o M et al. *Nature* 1997;390:45 (PMID 9363890) — original discovery of Klotho as anti-ageing gene
 Kurosu H et al. *Science* 2005;309:1829 (PMID 16123266) — Klotho extends lifespan in mice
 Drew DA et al. *JAMA* 2017;317:846 — circulating Klotho in CKD [not indexed]

GRADE C

MITOCHONDRIAL-DERIVED PEPTIDE

MOTS-c

Research only

HISTORY

Discovered in 2015 by Cohen and colleagues at USC; encoded within the 12S rRNA region of mitochondrial DNA.

SEQUENCE MRWQEMGYIFYPRKLR

16 residues

MECHANISM

Activates AMPK and modulates folate-methionine cycle; improves insulin sensitivity and mitochondrial function in preclinical models.

INDICATIONS

- None approved
- Investigational for metabolic dysfunction

DOSING

SC weekly total 5–15 mg/week, split into 2–3 doses (e.g., 5 mg M/W/F)

Microdosing (beginner) 3–5 mg SC twice weekly to assess tolerance

Cycling 4–8 wks on / equal off; max 4 cycles/year

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE · GRADE C RATES ENDOGENOUS MOTS-C BIOLOGY, NOT EXOGENOUS ADMINISTRATION

Animal studies show improved exercise capacity, insulin sensitivity, and lifespan in mice. No human clinical trials published as of 2026.

SAFETY

Limited human safety data · Anaphylaxis reported, especially with daily/high-frequency dosing · Theoretical concerns about modulating mitochondrial function long-term

ADVERSE REACTIONS

No human safety dataset; rodent work reports no overt toxicity at studied doses.

Clinical caveat. Evidence base: GRADE C rates endogenous MOTS-c biology — preclinical mitochondrial work plus human observational studies in which the intervention is exercise, heat, or metformin and MOTS-c is the measured analyte.

REGULATORY STATUS Research only · gray market · July 2026 PCAC review (obesity, osteoporosis)

REFERENCES

- Lee et al. Cell Metab 2015;21:443–54 (PMID 25738459)
Reynolds JC et al. Nat Commun 2021;12:470 — exercise and aging

GRADE C

SYNTHETIC DIPEPTIDE BIOREGULATOR

Vilon

Research chemical – no approved brand

HISTORY

Lys-Glu, from Vladimir Khavinson's “cytomedine” series at the St. Petersburg Institute of Bioregulation and Gerontology, studied almost entirely by that group.

SEQUENCE KE (Lys-Glu) — a dipeptide, not a fragment cut from a named parent
2 residues

MECHANISM

Proposed to enter cells and modulate tissue-specific gene expression; comitogenic in thymocyte culture with sphingomyelinase-pathway signalling.

INDICATIONS

- None established
- Research use only: thymic/immune gene-expression and geroprotection studies

DOSING

Human dosing None established — no published dose-finding study or validated regimen

Vendor “20 mg” vials A vial quantity, not a dose

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE · GRADE C RATES RUSSIAN-LANGUAGE CLINICAL REPORTS, NOT WESTERN TRIAL EVIDENCE

Rodent geroprotection and carcinogenesis work plus human lymphocyte and thymocyte cultures, nearly all from the Khavinson group.

SAFETY

No human safety data outside Russian-language reports · Chronic rodent dosing reported without adverse developmental effects

ADVERSE REACTIONS

Not characterized in humans. Long-term rodent dosing was reported without unfavorable developmental effects, from studies by the originating group.

Clinical caveat. Lys-Glu occurs inside both thymosin $\alpha 1$ and $\beta 4$, so vendors sell Vilon as a fragment of both.

REGULATORY STATUS Research chemical · no FDA approval · not on any 503A/503B bulks list · no registered trial

REFERENCES

- Khavinson VKh, Anisimov VN. Dokl Biol Sci 2000;372:261-3 (PMID: 10944717)
Sevostianova NN, Linkova NS, Polyakova VO, et al. Bull Exp Biol Med 2013;154:562-5 (PMID: 23486604)
Raikhlin NT, Bukaeva IA, Smirnova EA, et al. Bull Exp Biol Med 2004;137:588-91 (PMID: 15455093)

5-Amino-1MQ

Research only · gray market

HISTORY

Developed by the Sinclair laboratory and others as a selective inhibitor of nicotinamide N-methyltransferase (NNMT). Marketed within peptide communities as a 'metabolic peptide' despite being a small molecule. Never advanced to formal human clinical trials.

SEQUENCE 5-amino-1-methylquinolinium (small molecule, not a peptide)

Small molecule (~159 Da)

MECHANISM

Inhibits NNMT, the enzyme that methylates nicotinamide. NNMT inhibition raises intracellular NAD⁺ and lowers fat-storage gene expression in adipocytes...

INDICATIONS

- None approved
- Off-label/gray-market: weight loss, "metabolic health" enhancement

DOSING

Off-label oral 50–150 mg daily

Typical cycle 30-day on / 30-day off

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Strong rodent data (Sinclair lab and others) for reduced adiposity and improved metabolic markers in obese mice. No published human clinical trials.

SAFETY

No human safety data · NNMT involved in stem-cell regulation — chronic inhibition effects unknown · Theoretical effects on methyl-donor balance

ADVERSE REACTIONS

No published human safety data. Anecdotal reports of GI upset and mild headache. Long-term effects of NNMT inhibition uncharacterized in humans.

Clinical caveat. A small molecule, not a peptide — included here because it is marketed and sold through the same gray-market channels as injectable peptides. NNMT inhibition has compelling preclinical metabolic data but zero human clinical trials. Trial stage: preclinical only (rodent metabolic studies).

REGULATORY STATUS Research only · gray market (small molecule, not approved)

REFERENCES

- Neelakantan et al. *Biochem Pharmacol* 2017;147:141-152 (PMID 29155147)
Kraus et al. *Nature* 2014;508:258-62 (PMID 24717514)

FOXO4-DRI

FOX-04 DRI · research compound (senolytic)

HISTORY

Designed by Peter de Keizer's lab at University Medical Center Utrecht; published in Cell (Baar et al., 2017).

SEQUENCE d-LTLRKEPASEIAQSILEAYHSNSRGD

~46-residue D-retro-inverso construct

MECHANISM

In senescent cells, FOXO4 sequesters p53 in the nucleus, preventing apoptosis. The DRI peptide disrupts this interaction selectively in senescent cells (which express high FOXO4), releasing p53 to trigger apoptosis.

INDICATIONS

- None approved
- Investigational: senolytic therapy for aging-related dysfunction, frailty, sarcopenia, fibrotic disease
- Off-label/gray-market: longevity-clinic use without human evidence

DOSING

Investigational IV

100–400 µg/kg IV daily for 5-day cycles (gray-market protocol based on Baar et al. mouse dosing)

Doses as studied in the trials or protocols named; not an approved label for general use.

EVIDENCE

One landmark mouse study (Baar et al., Cell 2017). No published human trials. Mouse-to-human translation for senolytics has been poor — dasatinib + quercetin...

SAFETY

No human safety data · Theoretical risks if normal cells are inadvertently targeted · Cost-prohibitive for routine use (\$1000s per cycle in gray market)

ADVERSE REACTIONS

Not characterized in humans. Mouse studies showed transient fur loss during dosing cycles. Theoretical risk of immune-cell or stem-cell depletion...

Clinical caveat. One of the most-hyped longevity peptides with virtually no human evidence. The mouse data is striking but senolytic translation has been disappointing in humans. Gray-market use at \$1000s/cycle without human safety data carries unjustifiable risk for the strength of available evidence.

REGULATORY STATUS Research only · no human trials · not approved anywhere

REFERENCES

Baar MP et al. Cell 2017;169:132 (PMID 28340339) — original senolytic mouse model

G-Klotho Frag 33

Research compound · marketed fragment

HISTORY

A short synthetic fragment marketed as a klotho mimetic.

SEQUENCE Proprietary 33-residue fragment (sequences vary by vendor)

33 residues

MECHANISM

Proposed: recapitulates a subset of Klotho actions — TGF- β receptor 2 binding (anti-fibrotic), restoration of endogenous Klotho expression via miR-223-3p/lncRNA-TUG1 pathway, Nrf2 antioxidant activation.

INDICATIONS

- None approved
- Off-label/gray-market: anti-aging, renal protection claims, neuroprotection

DOSING

Off-label SC 0.5–1 mg twice weekly (vendor-suggested; no validated protocol)

Note Cycling commonly recommended by vendors despite no clinical basis

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

No published trials of "G-Klotho Frag 33" specifically. Adjacent research on KP1...

SAFETY

No human safety data · Vendor sequence and purity unverified

ADVERSE REACTIONS

Not characterized. Injection-site reactions, hypersensitivity, and contamination-related effects are the practical risks given uncertain manufacturing quality.

Clinical caveat. There is no published evidence for this fragment in any species. It is sold on the strength of the klotho literature, which studies a 130 kDa protein rather than a short peptide.

REGULATORY STATUS Research only · gray market · vendor-defined product

REFERENCES

- Yuan Q et al. Nat Commun 2022;13:438 — KP1 (Klotho-derived peptide 1) protects against kidney fibrosis via TGF- β R2
- Zhang et al. Theranostics 2024;14:420–435 (PMID 38164143)
- Doleckova I et al. Contipro (2017) — Klotho peptide fragment for topical cosmetic use; in vivo wrinkle reduction [not indexed]

GRADE D

MITOCHONDRIAL-DERIVED NEUROPROTECTIVE PEPTIDE

Humanin

HN · research compound

HISTORY

Discovered in 2001 by Hashimoto, Niikura, and Nishimoto (PNAS 2001;98:6336) in cDNA from surviving Alzheimer neurons.

SEQUENCE MAPRGFSCLLLLTSEIDLVPKRRRA (HNG: S14G variant)

24 residues (cytoplasmic translation) or 21 residues (mitochondrial)

MECHANISM

Anti-apoptotic: neutralizes pro-apoptotic Bcl-2 proteins (Bax, tBID). Activates pro-survival signaling via GP130/WSX-1/CNTFR receptor complex...

INDICATIONS

- None approved
- Investigational: Alzheimer's, T2D, ischemic injury; gray-market for neuroprotection/anti-aging

DOSING

Off-label SC

0.25–1 mg daily or every-other-day; S14G-HNG variant used at much lower doses (~1000× potency)

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Extensive preclinical literature — anti-apoptotic in neurons (Hashimoto 2001), beta-cell protection (Muzumdar 2009), cardioprotection, lifespan extension in *C. elegans*.

SAFETY

No human safety data for exogenous administration · Anti-apoptotic mechanism raises theoretical oncologic concern with chronic dosing

ADVERSE REACTIONS

Not characterized in humans. Animal studies have not shown major signals. Theoretical concern: pan-anti-apoptotic action could in...

Clinical caveat. Evidence base: extensive preclinical anti-apoptotic and lifespan literature + observational human studies of endogenous levels — no human RCTs of exogenous administration.

REGULATORY STATUS Research only · gray market · unapproved

REFERENCES

Hashimoto et al. Proc Natl Acad Sci U S A 2001;98:6336–41 (PMID 11371646)

Hashimoto et al. J Neurosci 2001;21:9235–45 (PMID 11717357)

Muzumdar RH et al. PLoS One 2009;4:e6334 — humanin improves insulin sensitivity, type 2 diabetes model

PART

Neurological

Nootropic and neurotrophic peptides, mostly outside US approval.

IN THIS PART

- Cerebrolysin
- Pinealon
- Selank
- Semax
- Dihexa
- DSIP (Delta Sleep-Inducing Peptide)
- PE-22-28

Cerebrolysin

Cerebrolysin (Ever Pharma)

HISTORY

Manufactured by EVER Neuro Pharma (Unterach, Austria) since the 1970s.

SEQUENCE Mixture of low-molecular-weight peptides (<10 kDa) and amino acids from enzymatic breakdown of porcine brain proteins

Peptide mixture (heterogeneous)

MECHANISM

Proposed to mimic the action of endogenous neurotrophic factors (BDNF, NGF, GDNF). Demonstrated effects on neurite outgrowth, neurogenesis...

INDICATIONS

- Stroke recovery (in approved countries)
- Vascular dementia and Alzheimer disease (adjunct)
- Traumatic brain injury

DOSING

IV 10–30 mL daily/EOD in 10–20 day courses; 20–50 mL daily for 7–14 days in acute stroke or TBI

SC 1 mL SC daily for 40–80 days (outpatient alternative)

EVIDENCE

Alvarez 2011 (n=133 moderate-AD subgroup of a dose-finding study): CIBIC+ improved across 10/30/60 mL doses at 24 weeks.

SAFETY

Generally well-tolerated · Hypersensitivity, vertigo, headache reported

ADVERSE REACTIONS

Most common: headache, vertigo, agitation, sweating, hyperhidrosis. Hypersensitivity reactions (rare). Injection-site irritation at IV catheter.

Clinical caveat. One of the very few peptide preparations with positive (if heterogeneous) RCT evidence in stroke recovery. The heterogeneous-mixture nature rather than a defined peptide complicates pharmacology — the exact active component remains unidentified.

REGULATORY STATUS *Approved in 50+ countries (notably China, Russia, Mexico, Austria) · not FDA-approved*

REFERENCES

- Heiss WD et al. Stroke 2012;43:630 (PMID 22282884) — pivotal stroke Phase 3
 Bornstein et al. Neurol Sci 2017;39:629–640 (PMID 29248999)
 Alvarez XA et al. Eur J Neurol 2011;18:59–68 (PMID 20500802) — moderate AD subgroup, n=133, dose-finding study

Pinealon

Research compound (Russia)

HISTORY

Developed by Vladimir Khavinson at the St. Petersburg Institute of Bioregulation and Gerontology. Part of the 'cytomedines' peptide family...

SEQUENCE EDR

3 residues

MECHANISM

Proposed to penetrate cells and bind specific gene promoters to regulate transcription, particularly of pineal-related and antioxidant genes.

INDICATIONS

- None Western-approved
- Russian use: cognitive disorders, age-related cognitive decline, neuroprotection
- Off-label / research: claimed effects on circadian rhythm via proposed pineal-axis gene regulation

DOSING

Typical Khavinson protocol 1–10 mg/day intranasal or SC

Course length 10–20 days, 2–3 courses per year

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

All published studies originate from the Khavinson group; not independently replicated outside Russia. In vitro evidence for antioxidant effects.

SAFETY

Limited Western safety data · Reportedly well-tolerated in Russian use · Long-term safety uncharacterized

ADVERSE REACTIONS

Generally well tolerated in Russian use. Reported: mild headache, occasional sleep changes. No serious AEs documented; long-term Western data lacking.

Clinical caveat. Evidence base rests entirely on Russian-group preclinical work (Khavinson cytomedine series) — no Western trials, no human RCTs published. One of several Khavinson-group 'cytomedine' peptides (alongside Epitalon and Vesugen) marketed for longevity and cognitive support.

REGULATORY STATUS Research only outside Russia

REFERENCES

Khavinson VKh group — Russian publications only [not indexed]

Selank

Approved in Russia

HISTORY

Synthetic analogue of tuftsin, developed at the Institute of Molecular Genetics, Moscow. Approved in Russia in 2009 as an anxiolytic and immune modulator.

SEQUENCE TKPRPGP

7 residues

MECHANISM

Proposed to modulate GABAergic and serotonergic systems, with effects on BDNF expression and immune function (tuftsin-like activity).

INDICATIONS

- Anxiety (Russia)
- Off-label: cognitive enhancement, immune modulation

DOSING

Intranasal 300–900 µg daily

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Russian RCTs reported in non-Western journals; few independent replications. Animal data is more consistent.

SAFETY

Generally well-tolerated in Russian use · Long-term Western safety data lacking · Limited drug-interaction data

ADVERSE REACTIONS

Generally well tolerated in Russian clinical use. Reported: occasional drowsiness, dry mouth, nasal irritation (intranasal route).

Clinical caveat. Evidence base: Russian preclinical anxiolytic studies + limited small clinical work — no Western RCTs published. Approved in Russia for anxiety; claimed to lack the dependence potential and sedation of benzodiazepines. No Western trials and limited drug-interaction data — efficacy claims rest entirely on non-Western evidence. Trial stage: Russian Phase 2/3 approved as intranasal Selank; no Western trials.

REGULATORY STATUS *Approved in Russia · research elsewhere*

REFERENCES

Russian publications (Inozemtsev, Semenova) — anxiolytic preclinical and small clinical studies [not indexed]

Semax

Approved in Russia

HISTORY

Synthetic heptapeptide derived from ACTH(4-10); developed at the Institute of Molecular Genetics, Moscow. Approved in Russia (1994 for stroke, 2003 for cognitive disorders). No Western regulatory approval.

SEQUENCE MEHFPGP

7 residues

MECHANISM

Proposed to upregulate BDNF expression and modulate dopamine/serotonin systems centrally. Devoid of corticotropic activity (lacks the N-terminal portion of ACTH).

INDICATIONS

- Stroke (acute), encephalopathy, cognitive disorders (Russia)
- Off-label cognitive enhancement

DOSING

Intranasal 600–1800 µg daily

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Russian trials for stroke recovery; quality of evidence and independent replication limited.

SAFETY

Generally well-tolerated in Russian clinical use · Long-term Western data lacking

ADVERSE REACTIONS

Generally well tolerated. Reported: nasal irritation (intranasal route), occasional mild anxiety or restlessness, rare reports of insomnia. Long-term safety in Western populations not characterized.

Clinical caveat. Evidence base: Russian preclinical and small clinical stroke/cognitive studies — no Western RCTs published. Approved in Russia for stroke and cognitive disorders. ACTH(4-10)-derived but lacks corticotropic activity. Widely used off-label as a nootropic without supporting Western trial data. Trial stage: Russian Phase 2/3 approved as intranasal Semax; no Western trials.

REGULATORY STATUS *Approved in Russia · research elsewhere · July 2026 PCAC review (cerebral ischemia, migraine, trigeminal neuralgia)*

REFERENCES

Russian publications (Myasoedov group) — stroke, cognitive enhancement [not indexed]

Dihexa

Research only · gray market

HISTORY

Developed by John Harding's laboratory at Washington State University as a cognitive enhancer. Modified from angiotensin IV (AT4) with...

SEQUENCE N-hexanoyl-Tyr-Ile-(6) amino hexanoic amide

2 residues (*heavily modified*)

MECHANISM

Hepatocyte growth factor (HGF) / c-Met signaling activator. Stimulates dendritic arborization, synaptogenesis, and neural connectivity. Reported ~10 million times more potent than BDNF in some neurite-outgrowth assays.

INDICATIONS

- None approved
- Off-label/gray-market: nootropic, cognitive enhancement, investigational for neurodegenerative disease

DOSING

Off-label oral or sublingual 5–30 mg/day

Protocols vary widely No standardized regimen

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Strong rodent data for cognitive improvement in MCI and Alzheimer's disease models (Wright and Harding laboratories). No human clinical trials published.

SAFETY

Limited safety data · c-Met signaling activation raises theoretical oncogenic concerns at chronic dosing · Long-term effects unknown

ADVERSE REACTIONS

Limited human data. Reported by users: mild headache, transient fatigue, occasional anxiety or insomnia at higher doses. No serious AEs documented in animal studies.

Clinical caveat. One of the few peptide-derived nootropics with truly compelling animal data — but zero human trial data. c-Met signaling activation raises theoretical oncogenic concerns at chronic dosing that have not been addressed in any human study. Trial stage: preclinical only (rodent cognitive studies).

REGULATORY STATUS Research only · gray market

REFERENCES

McCoy et al. J Pharmacol Exp Ther 2012;344:141-54 (PMID 23055539)

DSIP (Delta Sleep-Inducing Peptide)

Research only · gray market

HISTORY

Isolated by Schoenenberger and Monnier in 1977 from the cerebral venous blood of rabbits in slow-wave sleep. Promoted delta (slow-wave) sleep when administered to...

SEQUENCE WAGGDASGE

9 residues

MECHANISM

Endogenous nonapeptide with proposed effects on slow-wave sleep, stress-response modulation, and neuroendocrine regulation. No specific cognate receptor has been identified despite decades of investigation.

INDICATIONS

- None approved
- Off-label gray-market: insomnia, withdrawal syndromes (alcohol, opioid), chronic pain

DOSING

Off-label SC at bedtime 100–500 µg

Typical pattern Intermittent dosing, not nightly

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Small Russian and Eastern European trials in the 1980s suggested modest sleep-quality improvements. No modern RCT data. Mechanism remains poorly understood.

SAFETY

Generally well-tolerated in limited use · Long-term safety not characterized

ADVERSE REACTIONS

Limited human data. Reported: vivid dreams, mild morning grogginess at higher doses, injection-site reactions. No serious AEs documented.

Clinical caveat. One of the original 'designer sleep peptides' with intriguing 1970s–80s data that never translated into approved therapy. Modern sleep pharmacology (orexin antagonists) has more rigorously studied alternatives. Trial stage: small 1970s–80s clinical trials; never reached registration; no modern trials.

REGULATORY STATUS *Research only · gray market · July 2026 PCAC review (opioid withdrawal, chronic insomnia, narcolepsy)*

REFERENCES

Schoenenberger et al. Proc Natl Acad Sci U S A 1977;74:1282–6 (PMID 265572)

No modern clinical trials available [not indexed]

PE-22-28

Research compound (spadin analog)

HISTORY

Optimized fragment of spadin — a 17-residue cleavage product of sortilin propeptide that selectively inhibits TREK-1 potassium channels.

SEQUENCE Approximate: optimized 7-mer spadin fragment

7 residues

MECHANISM

Selectively inhibits TREK-1 (TWIK-related potassium channel 1), a background K⁺ channel expressed in serotonergic and cortical neurons. TREK-1 inhibition is hypothesized to produce antidepressant effects with...

INDICATIONS

- None approved
- Investigational: depression, treatment-resistant depression
- Off-label intranasal use in some longevity/wellness clinics

DOSING

Off-label intranasal 400 µg per spray, 2 sprays per nostril daily (≈1.6 mg/day total)

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Preclinical only — rodent models support antidepressant activity via TREK-1 inhibition. Mendelian-validation evidence: TREK-1 polymorphisms are associated with...

SAFETY

No human safety data · Theoretical cardiac concern (TREK-1 expressed in atrial myocytes) · Long-term effects unknown

ADVERSE REACTIONS

Not characterized in humans. Rodent studies have not shown significant adverse effects but the trial corpus is small.

Clinical caveat. Mechanistically interesting — TREK-1 is a validated antidepressant target — but PE-22-28 itself has no human clinical evidence. The intranasal route is convenient but unstudied; cardiac TREK-1 expression is a theoretical safety concern for chronic dosing. Trial stage: preclinical only.

REGULATORY STATUS Research only · no human trials

REFERENCES

Djillani et al. Front Pharmacol 2017;8:643 (PMID 28955242)

PART

Investigational Oncology

Early-stage oncology peptides, included for context.

IN THIS PART

- Met-5-Enkephalin
- PNC-27

Met-5-Enkephalin

Opioid Growth Factor · OGF

HISTORY

Identified by Hughes and Kosterlitz in 1975 as the first endogenous opioid peptide. Ian Zagon's lab subsequently characterized it as Opioid Growth Factor (OGF)...

SEQUENCE YGGFM

5 residues

MECHANISM

Activates OGF_r (opioid growth factor receptor, nuclear-shuttling), inducing p16 and p21 cyclin-dependent kinase inhibitors. Inhibits cell proliferation in OGF_r-expressing tumors and immune cells. Mechanistically distinct from classical opioid analgesia.

INDICATIONS

- None approved
- Investigational: pancreatic cancer (Phase 2, modest activity)
- Off-label/research: multiple sclerosis, Crohn disease (same axis as low-dose naltrexone)

DOSING

Investigational IV 10 mg/m² IV weekly (Zagon Phase 2 pancreatic cancer)

Off-label SC 10 mg SC twice daily, twice weekly

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Phase 2 in pancreatic cancer (Smith et al., Open Access Journal of Clinical Trials 2010) showed median survival ~8 months — disappointing relative to chemotherapy.

SAFETY

Generally well tolerated in trials · IV pulse causes mild flushing and hypotension · Long-term safety not characterized

ADVERSE REACTIONS

Flushing, fatigue, mild GI upset reported in trials. Theoretical concern about delayed wound healing given anti-proliferative effects on normal cells.

Clinical caveat. The OGF_r mechanism is well-characterized but met-enkephalin has a plasma half-life of minutes, making direct dosing inefficient. Low-dose naltrexone (LDN, 1.5–4.5 mg PO nightly) blocks OGF_r transiently...

REGULATORY STATUS Research only · not approved as cancer therapy

REFERENCES

Smith et al. Open Access J Clin Trials 2010;2010:37-48 (PMID 20890374)

PNC-27

Research only · gray market

HISTORY

Designed by Michael R. Pincus and colleagues. Fuses an HDM-2 (MDM2) binding domain from p53 with a membrane-translocating sequence.

SEQUENCE PPLSQETFSDLWKLKKWKMRNQFWVKVQRG

32 residues

MECHANISM

Targets HDM-2 (MDM2) expressed on cancer-cell plasma membranes. The peptide forms pores in tumor-cell membranes (cell-selective necrosis/lysis), sparing normal cells which lack surface HDM-2.

INDICATIONS

- None approved
- Investigational/off-label: various cancers (forum and clinic use)

DOSING

Off-label SC 200–300 µg SC three times weekly (gray-market protocol)

Reported off-label or gray-market practice, recorded for history-taking. Not recommended doses; no approved label supports them for this use.

EVIDENCE

Preclinical only — multiple cell-line and animal studies. No published Phase 1 human safety data. Clinical efficacy claims rest entirely on in vitro and rodent work.

SAFETY

No human safety profile characterized · Off-label injectable use carries undefined risks · Theoretical immunogenicity from 32-residue synthetic peptide

ADVERSE REACTIONS

No characterized human safety data. Gray-market reports describe injection-site reactions, fatigue, fevers. Tumor-lysis-style syndromes are theoretically possible but not documented.

Clinical caveat. Often marketed online as a 'cancer cure' — there is no human clinical evidence to support this claim. Compelling mechanism and preclinical data have not yet translated to human trials. Use outside an IRB-approved protocol is not supported by the evidence. Trial stage: preclinical only; no human trials.

REGULATORY STATUS Research only · not approved anywhere · no human trial data

REFERENCES

Michl et al. Int J Cancer 2006;119:1577–85 (PMID 16688716)
No human trials published [not indexed]

PART

GI & Other

Gut-active and antimicrobial peptides.

IN THIS PART

- Linaclotide
- LL-37 (Cathelicidin)
- Larazotide acetate

Linaclotide

Linzess · Constella

HISTORY

Developed by Ironwood Pharmaceuticals; FDA-approved 2012 for IBS-C and chronic idiopathic constipation. Pediatric approval added 2023.

SEQUENCE CCEYCCNPACTGCGY (with three intramolecular disulfides)

14 residues

MECHANISM

Activates intestinal guanylate cyclase-C (GC-C) on the luminal surface of enterocytes; raises intracellular cGMP, opens CFTR, drives chloride and water secretion into the intestinal lumen. Local action; minimal systemic absorption.

INDICATIONS

- IBS with constipation (IBS-C)
- Chronic idiopathic constipation (CIC)
- Functional constipation in pediatric patients (ages 6+)

DOSING

Oral daily (CIC) 145 µg once daily

Oral daily (IBS-C) 290 µg once daily

Pediatric FC (6–17 yrs) 72 µg once daily

EVIDENCE

Multiple pivotal Phase 3 trials show significant improvement in stool frequency, abdominal pain, and global IBS-C symptom scores.

SAFETY

Diarrhea is the main dose-limiting effect · Contraindicated in pediatric patients < 2 years (lethal dehydration in juvenile mice) · GI obstruction must be ruled out

ADVERSE REACTIONS

Most common: diarrhea (16–20%, dose-limiting), abdominal pain (7%), flatulence, abdominal distension, viral gastroenteritis, headache. Severe diarrhea may require dose interruption. Boxed warning for fatal dehydration in pediatric patients...

Clinical caveat. Take 30 minutes before the first meal of the day for best effect — food substantially reduces absorption. Diarrhea is dose-limiting; consider starting with the lower 145 µg dose even for IBS-C in sensitive patients.

REGULATORY STATUS FDA-approved (Linzess US 2012, Constella EU) · EMA

REFERENCES

Leombo et al. N Engl J Med 2011;365:527-36 (PMID 21830967)

Chey et al. Am J Gastroenterol 2012;107:1702-12 (PMID 22986437)

LL-37 (Cathelicidin)

Research only

HISTORY

The mature 37-amino-acid antimicrobial peptide derived from the human cathelicidin gene hCAP18. Studied since the 1990s for antimicrobial...

SEQUENCE LLGDFFRKSKEKIGKEFKRIVQRIKDFLRNLPRTES

37 residues

MECHANISM

Direct antimicrobial via membrane disruption (broad-spectrum). Also modulates innate immune cell function, wound healing...

INDICATIONS

- None approved
- Investigational: chronic wounds, infections, autoimmune skin disease (where its dysregulation is pathogenic)

DOSING

Investigational topical No standardized human protocol

Doses as studied in the trials or protocols named; not an approved label for general use.

EVIDENCE

Two randomized placebo-controlled trials have evaluated topical LL-37 in hard-to-heal venous leg ulcers. A 34-patient first-in-human dose-ranging trial found...

SAFETY

Limited human safety data · Pro-inflammatory role in some skin diseases · Cytotoxicity at higher concentrations

ADVERSE REACTIONS

Limited human data. Topical or trial use: skin irritation, erythema. Pro-inflammatory role implicated in rosacea, psoriasis...

Clinical caveat. Evidence base: two randomized placebo-controlled wound trials, the smaller positive at low dose and the larger Phase IIb negative in its full cohort with only a post hoc subgroup signal.

REGULATORY STATUS *Research only · investigational topical · no human therapeutic approval*

REFERENCES

Zanetti M et al. FEBS Lett 1995;374:1 — cathelicidin family review

Grönberg A et al. Wound Repair Regen 2015;22:613-21 (PMID 25041740) — first-in-human randomized dose-ranging, venous leg ulcers

Mahlapuu M et al. Wound Repair Regen 2022;29:938-950 (PMID 34687253) — HEAL LL-37 Phase IIb randomized placebo-controlled, negative in full cohort

Larazotide acetate

Investigational (9 Meters Biopharma)

HISTORY

Synthetic 8-amino-acid peptide developed as a 'leaky gut' modulator; investigated as adjunct to gluten-free diet in celiac disease. Phase 3 CeDLara trial (2022) failed primary endpoint; clinical development largely halted.

SEQUENCE GGVLVQPG

8 residues

MECHANISM

Antagonizes zonulin, the endogenous regulator of intestinal tight junctions; reduces gluten-induced permeability and downstream inflammation.

INDICATIONS

- Investigational for celiac disease
- Off-label gray-market use for "leaky gut" complaints

DOSING

Trial protocols 0.5–1 mg orally three times daily before meals

Doses as studied in the trials or protocols named; not an approved label for general use.

EVIDENCE

Earlier Phase 2b trials showed reduced symptoms vs placebo, but pivotal Phase 3 CeDLara failed.

SAFETY

Headache common · Generally well-tolerated · Long-term safety not established

ADVERSE REACTIONS

Most common in trials: headache (12%), upper respiratory tract infection (10%), urinary tract infection, fatigue. Generally well tolerated; failed Phase 3 on efficacy rather than safety grounds.

Clinical caveat. Evidence base: pivotal Phase 3 CeDLara failed its primary endpoint (2022); off-label "leaky gut" use rests on mechanistic plausibility and anecdote, not RCT evidence. The pivotal Phase 3 CeDLara trial (2022) failed its primary endpoint. Off-label "leaky gut" marketing substantially exceeds the actual evidence base.

REGULATORY STATUS *Investigational · not approved*

REFERENCES

Leffler DA et al. Gastroenterology 2015;148:1311 (PMID 25683116) — celiac CeD-PRO Phase 2b

Clinical pearls

A prescribing framework, the order of operations before reaching for a peptide, and the conversation a Tier 3 molecule requires.

A THREE-TIER PRESCRIBING FRAMEWORK

Tier 1 · approved, on-label	Highest evidence; the default when an approved agent fits the need. Semaglutide for type 2 diabetes, tirzepatide for obesity, tesamorelin for HIV-associated lipodystrophy.
Tier 2 · approved, off-label	Lawful. Document the rationale and obtain informed consent. Tesamorelin for non-HIV visceral adiposity; sermorelin in adults outside approved pediatric use.
Tier 3 · not approved	Compounded or research-only. Elevated risk. This is not off-label prescribing: there is no approved label to depart from. BPC-157, TB-500, CJC-1295, ipamorelin, GHK-Cu, Vilon.

FOUNDATIONS BEFORE PEPTIDES

- Optimize the substrate first. Peptides amplify a healthy foundation — diet, sleep, hydration, training — they do not substitute for one.
- Start low, go slow. Begin at the lowest plausible dose and titrate on response.
- One variable at a time. Stacking several agents at once makes an adverse effect impossible to attribute.
- Set a stopping rule before starting, and a date to reassess.

THE TIER 3 CONVERSATION

1. Name the article and its status plainly: not approved for this use, or not approved at all.
2. State what is known and what is not, including that the evidence may be animal or in vitro only.
3. Name the approved alternatives, including doing nothing.
4. Record the monitoring plan and the threshold to stop.

Cycling, stacking, and monitoring

Cycling, stacking, and the practical rules that follow from the three-tier framework.

PEARL 5: CYCLING PATTERNS REPORTED IN WELLNESS PRACTICE

Commonly reported: 8-12 weeks on, 4-8 weeks off. No comparative trial validates these intervals, so treat them as practice patterns rather than dosing guidance. The defensible part is the habit: reassess clinical endpoints at each interval's end rather than perpetuating open-ended prescriptions.

PEARL 6: READ THE CURRENT 503A ROSTER, NOT A CACHED ONE

FDA's 503A interim-policy roster, updated 14 May 2026, holds six substances in Category 2 as raising significant safety risks. The peptide-class entries are ibutamoren mesylate and kisspeptin-10. Most peptides once in Category 2 now sit on a separate table of nominations withdrawn by their nominators, in no category at all:

- BPC-157, TB-500, CJC-1295, AOD-9604, thymosin alpha-1, KPV, MOTS-c, melanotan II, semax, epitalon and emideltide (DSIP) — nominations withdrawn.
- Ipamorelin — withdrawn from 503A Category 2; still 503B Category 2.
- GHRP-2 and GHRP-6 — 503A Category 3, though 503B Category 2. The section matters: printing "Category 2" unqualified is ambiguous.
- GHK-Cu — Category 1 for non-injectable routes after a nominator withdrew only the injectable nomination.
- Sermorelin — in none of the three categories.

A withdrawn nomination is not clearance, and Category 1 means a substance is not on the Bulks List at all. FDA does not intend to categorize substances nominated on or after 7 January 2025, so the scheme is closed to new entrants. Seven peptides went to the Pharmacy Compounding Advisory Committee on 23-24 July 2026 — BPC-157, KPV, TB-500, MOTS-c, emideltide, epitalon and semax — with FDA staff concluding against listing each one. Check the roster rather than a cached figure before compounding anything.

PEARL 7: RECOGNIZE RED FLAGS AND KNOW WHEN TO STOP

GLP-1	Severe abdominal pain (pancreatitis), neck mass (MTC workup), hypersensitivity
GH secretagogues	IGF-1 above ULN, acromegaloïd features
Compounded	Fever, injection-site infection, LFT/RFT derangement

CYCLING, STACKING, AND MONITORING • CONTINUED

PEARL 8: DOCUMENT INFORMED CONSENT IN DETAIL FOR OFF-LABEL AND GRAY-MARKET USE

Off-label use of FDA-approved drugs is legal; gray-market peptides carry elevated risk.

Document the following:

- GRADE of evidence
- Regulatory status
- Alternatives offered
- Expected effects
- Monitoring plan
- Discontinuation criteria

Verbal-only consent is professionally vulnerable; written documentation protects clinician and patient.

PEARL 9: TREAT PREGNANCY AND LACTATION AS CONTRAINDICATIONS BY DEFAULT

Most peptides have no FDA pregnancy category; almost none have lactation data. Discontinue at conception planning for non-essential peptides.

Exceptions are specifically labeled (e.g., insulin is mandatory in pregnancy; teriparatide is contraindicated). Counsel reproductive-age patients before initiation.

CYCLING, STACKING, AND MONITORING • CONTINUED

PEARL 10: SET THE BASELINE BEFORE STARTING — AND TRACK OUTCOMES

GLP-1 RAs	Weight/waist
GH secretagogues	IGF-1
Melanocortins	Validated libido scales
Mitochondrial peptides	Energy/sleep symptom scores

Re-check at 8-12 weeks. If no measurable change against baseline, stop the drug rather than escalating dose.

PEARL 11: DOSE GH SECRETAGOGUES AT NIGHT, ON AN EMPTY STOMACH

Best taken at night, 2-3 h after the last meal, on an empty stomach — this synergizes with the natural GH surge during sleep. Counsel patients that body-composition effects appear gradually, over 3-6 months.

Stop and investigate immediately for: severe abdominal pain (possible pancreatitis) or a neck mass (possible medullary thyroid carcinoma) on GLP-1 therapy, IGF-1 above the ULN or acromegaloïd features on GH secretagogues, or fever/injection-site infection/LFT-RFT derangement with compounded peptides.

Neuropsychiatric and addiction signals in GLP-1 therapy

A cross-class summary of the signals reported across the incretin agents.

A cross-class summary, collected once rather than repeated in each entry. All signals below are hypothesis-generating; none is an approved indication.

WHY A REWARD-PATHWAY SIGNAL IS PLAUSIBLE

GLP-1 receptors are expressed in mesolimbic reward circuitry — the ventral tegmental area and nucleus accumbens — where activation modulates dopaminergic signaling. This gives a mechanistic rationale for the recurring observation that GLP-1 receptor agonists may reduce alcohol and nicotine consumption and blunt some reward-driven behaviors. Plausibility is not efficacy: the human evidence remains early.

ALCOHOL USE DISORDER (AUD)

- Semaglutide: the strongest prospective signal, now in dedicated trials (e.g., SEMALCO) after observational and early clinical data suggested reduced intake.
- Tirzepatide: shows an AUD-reduction signal exceeding other agents in observational datasets, but heavily confounded.
- Dulaglutide: linked to reduced alcohol use during a smoking-cessation study (Probst et al., JCI Insight 2023).
- Liraglutide and exenatide: show preclinical/observational signals.

SMOKING CESSATION

The clearest randomized signal is for exenatide: one positive RCT in combination with a nicotine patch. Other agents have observational or preclinical data only.

INTERPRETATION FOR PRESCRIBERS

None of these is an approved indication, and none should drive drug selection today. The class-wide consistency is worth knowing — especially when a patient with obesity or T2DM also has AUD or tobacco dependence — but dedicated Phase 3 trials are still in progress.

Treat any addiction benefit as a possible secondary observation, not a reason to prescribe.

Receptor desensitization and peptide cycling

Why many peptides lose effect with continuous use, and what follows for dosing.

Why many peptides lose effect with continuous use, and the rationale offered for scheduled off-cycles ("peptide holidays"). The mechanism is well characterised; the specific schedules are practice convention.

Many peptide effects fade with sustained or frequent dosing (tachyphylaxis), usually from adaptive down-regulation of the target receptor — sometimes compounded by counter-regulatory feedback or anti-drug antibodies. Matching dosing to the body's natural rhythm, and the rationale offered for scheduled off-cycles is that receptor responsiveness may recover during the interval.

HOW DESENSITIZATION HAPPENS

Receptor uncoupling & internalization

Agonist binding recruits GPCR kinases and β -arrestin, which uncouple the receptor from its G-protein and draw it into the cell (endocytosis); sustained signaling then lowers receptor number (down-regulation).

Exposure pattern

Pulsatile, physiologic signaling preserves sensitivity; continuous or supraphysiologic exposure (long-acting analogues, frequent high doses) desensitizes fastest — hence pulsatile GH secretagogues that mimic the overnight rhythm hold up better.

Counter-regulatory feedback

The body opposes sustained stimulation — e.g., somatostatin tone rises against continuous GH-axis drive — blunting response independent of the receptor.

Anti-drug antibodies

Repeated exposure to a foreign peptide can raise neutralizing antibodies that cut efficacy or, rarely, cause hypersensitivity — likelier with impure gray-market product.

RECEPTOR DESENSITIZATION AND PEPTIDE CYCLING · CONTINUED

CLASS EXAMPLES

GH secretagogues

Hexarelin desensitizes fastest — GH responses can fade within 2–4 wks of daily use; ipamorelin and CJC-1295 are reported to fade more slowly. Off-cycle intervals of roughly 2–4 wks are described in wellness practice, not established by trial.

GnRH agonists (leuprolide, goserelin)

Desensitization is the therapeutic goal — continuous dosing down-regulates pituitary GnRH receptors, suppressing LH/FSH after an initial flare.

Melanocortins and others show variable tachyphylaxis, arguing for on-demand or spaced use.

WHY PEPTIDE HOLIDAYS

The proposed rationale is that off-cycles may allow receptor density and signalling to recover, and may limit antibody formation.

- Commonly reported: 8–12 wks on, 4 wks off
- Shorter intervals described for fast-desensitizing agents (hexarelin ≈4 wks on / 2–4 wks off)

A waning effect is a reason to reassess rather than escalate the dose. These schedules are not validated by comparative clinical trials and are not evidence-based dosing recommendations.

Lab & monitoring reference

What to draw before starting, what to follow during therapy, and the values that mean stop rather than titrate.

BASELINE PANEL

- Comprehensive metabolic panel, fasting glucose, HbA1c, fasting lipid panel
- IGF-1 for any GH secretagogue or GHRH analogue
- Thyroid function (TSH, free T4), complete blood count, hs-CRP
- Fasting insulin where a metabolic-syndrome workup is indicated
- Sex hormones (total and free testosterone, estradiol, DHEA-S) where relevant
- For GLP-1 candidates: personal and family history of medullary thyroid carcinoma or MEN2 (contraindication), and diabetic retinopathy screening in type 2 diabetes

MONITORING DURING THERAPY

GLP-1 and incretin agents	Every 3 months: HbA1c, fasting glucose, lipids, hepatic and renal function. Screen for pancreatitis, gallbladder disease, and retinopathy progression.
GH secretagogues, GHRH analogues	IGF-1 at 6-8 weeks then every 3-6 months, targeting the mid-normal age-adjusted range. Fasting glucose and HbA1c: GH lowers insulin sensitivity.
Melanocortin agonists	Full-body skin examination at baseline and periodically; blood pressure.
Research-only molecules	No validated monitoring exists. Follow the organ systems the mechanism implicates and document that no established protocol was available.

An IGF-1 above the age-adjusted reference range on a secretagogue is a stopping signal, not a titration target.

Anti-doping urine and plasma testing

What gets tested, where, and what cannot be detected.

DIRECT VS INDIRECT DETECTION

Exogenous synthetic peptides (GHRPs, GHRH analogues, melanocortin agonists, BPC-157, TB-500) are detected directly by LC-MS/MS in urine or plasma.

Peptides identical to endogenous hormones (somatropin, insulin, IGF-1) cannot be distinguished by identity alone; detection uses isoform analysis and indirect biomarker panels.

VALIDATED URINE ASSAYS (WADA-RECOGNIZED)

GHRPs (GHRP-2, GHRP-6, hexarelin, ipamorelin)	intact peptide and metabolites; plasma $t_{1/2}$ 15–30 min; urine detection 24–72 h
GHRH analogues (sermorelin, CJC-1295 no-DAC, tesamorelin)	intact peptide; plasma $t_{1/2}$ 7–30 min, urine 1–3 d. CJC-1295 with DAC has plasma $t_{1/2}$ 5–8 d (urine 2–3 weeks)
BPC-157	validated urine metabolite assay (2024); plasma $t_{1/2}$ ~30 min; urine 24–48 h
TB-500 / thymosin β4	urine peptide and metabolites; plasma $t_{1/2}$ 1–3 d; urine 5–7 d
Insulin analogues	analogue-specific signatures (B30 modifications, glargine A21G) by LC-MS/MS in plasma

ANTI-DOPING URINE AND PLASMA TESTING • CONTINUED

INDIRECT BIOMARKER PANELS (WHEN DIRECT DETECTION FAILS)	
Recombinant GH (somatropin)	identical to natural 22-kDa hGH; direct detection fails. WADA uses an isoform ratio (rhGH suppresses 20-kDa GH) and the GH-2000 score (IGF-1 + P-III-NP); detection windows ~24–48 h (isoform) and up to ~2 weeks (biomarker)
IGF-1 / IGF-1 LR3	short bound half-life (12–15 h); LR3 at low doses is hard to catch; integrated into the GH-2000 score
EPO and ESAs	IEF/SDS-PAGE band pattern plus the Athlete Biological Passport (reticulocyte %, OFF-score)

WHAT CANNOT BE RELIABLY TESTED - MATRIX & STABILITY

Endogenous-identical biologics outside the somatropin / insulin / IGF-1 indirect-detection framework; niche peptides without validated metabolite signatures (DSIP, dihexa, several Khavinson bioregulators, PE22-28) — no anti-doping assay exists.

Urine is the primary WADA matrix (non-invasive, long window); plasma / DBS is preferred for short- $t_{1/2}$ peptides, biomarker panels, and the somatropin isoform assay.

Urine peptides degrade rapidly at room temperature — cold-chain transport and $-20^{\circ}\text{C}/-80^{\circ}\text{C}$ storage are required.

Adverse effects and overdose

Class-based adverse-effect profiles, and a structured response to excess.

Class-based adverse-effect profiles, GH-excess recognition and management, and overdose workup for compounded and gray-market peptide use.

GENERAL RISKS (ALL UNAUTHORIZED PEPTIDES)

- Injection-site infection from non-sterile preparation or technique.
- Endotoxin contamination — 8% of tested samples in one analysis (fever, chills, septic-shock risk).
- Hypersensitivity and GI effects, plus contamination with lead or wrong active ingredient — falsified products have carried arsenic up to 10x ICH limits.
- Skin hyperpigmentation and new or darkening moles — from melanocyte overstimulation (melanocortin-class peptides) or as a sign of heavy-metal toxicity; chronic arsenic exposure classically causes diffuse hyperpigmentation and skin lesions.
- New or changing pigmented lesions warrant dermatologic evaluation.

Endotoxin contamination has been found in up to 8% of tested unauthorized-peptide samples and can cause fever, chills, and septic shock — sterile sourcing and technique are critical.

GH SECRETAGOGUES (CJC-1295, IPAMORELIN, TESAMORELIN)

- Fluid retention — edema, arthralgia, carpal-tunnel syndrome.
- Glucose intolerance and elevated IGF-1, with theoretical malignancy risk from sustained GH/IGF-1 (detailed stats and monitoring in GH-excess section).
- Injection-site reactions, paresthesia, myalgia.

IGF-1 / IGF-1 LR3

- Hypoglycemia — LR3 is 2-3x more potent than native IGF-1 acutely, with 4-8x greater cumulative suppression over 4 h; IGF-1 also suppresses glucagon, impairing recovery.
- Intracranial hypertension, slipped capital femoral epiphysis, scoliosis progression (especially adolescents); lymphoid-tissue hypertrophy.

ADVERSE EFFECTS AND OVERDOSE • CONTINUED**GHK-CU (COPPER PEPTIDE)**

- Low cytotoxicity preclinically; at overdose, copper toxicity dominates — nausea, vomiting, acute renal failure, hemolytic anemia, hepatic injury.
- FDA-approved copper histidinate (Zycubo) carries renal-tubular, liver, and hematologic warnings.
- Hypothetical zinc deficiency with chronic or high-dose use — copper and zinc compete for intestinal absorption, so sustained copper loading could in theory lower zinc status.
- Not documented at cosmetic topical doses; consider monitoring copper and zinc with prolonged systemic use.

BPC-157

- Good preclinical safety (LD50 not reached in mice, rats, rabbits, dogs); no human safety data.
- Injection-site infection and compartment syndrome (requiring fasciotomy) reported after cocktail injection.

Recognizing and diagnosing growth-hormone excess from secretagogue and GHRH-analogue use.

Sustained GH/IGF-1 activation from off-label GHRH analogues (sermorelin, CJC-1295, tesamorelin) and secretagogues (ipamorelin, hexarelin, MK-677) can produce a partial, biochemical, or -- rarely -- clinically apparent acromegaloid phenotype carrying cardiovascular, metabolic, and oncologic risk.

Many patients present after months to years of compounded use without ever having had IGF-1 measured.

ADVERSE EFFECTS AND OVERDOSE • CONTINUED**CLINICAL SPECTRUM**

Functional excess (most common)	Fluid retention, edema, arthralgia, carpal-tunnel symptoms, paresthesia, sleep disturbance; largely reversible off-drug.
Acromegaloid features (prolonged/supraphysiologic use)	Soft-tissue swelling of hands and feet (ring or shoe tightening is often the first clue), facial coarsening, prognathism, macroglossia, skin-tag proliferation, hyperhidrosis; may be irreversible.
Metabolic	Impaired glucose tolerance, new-onset T2DM (tesamorelin HR 3.3 vs placebo), insulin resistance, dyslipidemia.
Cardiovascular	Hypertension, LV hypertrophy, diastolic dysfunction, OSA.
Oncologic concern	Sustained IGF-1 theoretically promotes colorectal, breast, and prostate proliferation; no causal data in secretagogue users, but the association in endogenous acromegaly is well-established.

WHEN TO SUSPECT & HOW TO WORK UP

Suspect in any GH-axis-peptide user with: new arthralgia, paresthesia, ring-size change, fluid retention, new T2DM/OSA/hypertension, asymmetric soft-tissue change (hands/feet/jaw), or an IGF-1 above the age-adjusted ULN on monitoring.

Serum IGF-1 (age/sex-adjusted)	The workhorse; single random draw, unaffected by pulsatile GH; repeat in 2-4 wks if equivocal.
OGTT with GH suppression	75 g, GH at 0-120 min; normal nadir < 1 microg/L (< 0.4 on ultrasensitive assays). Confirms excess but doesn't distinguish exogenous from endogenous.
Pituitary MRI	If IGF-1 stays elevated > 8 wks after stopping, or biochemistry is severe, to exclude a somatotroph adenoma.
Ancillary	Fasting glucose/HbA1c, lipids, ECG, echo if cardiac symptoms, colonoscopy if age > 50, sleep study if OSA features.

IGF-1 targets, dose reduction, and a structured workup for compounded-peptide overdose.

ADVERSE EFFECTS AND OVERDOSE • CONTINUED**IGF-1 TARGETS**

On therapy	Upper-half of the age-adjusted range, never above ULN.
Above ULN	Reduce dose or lengthen interval; recheck in 4-6 wks.
> 1.5x ULN	Discontinue; evaluate for acromegaloid features; consider OGTT-GH and pituitary MRI.

Tesamorelin — 47% of treated patients exceeded ULN at 26 wks in the pivotal trial; monitor at baseline, 4-6 wks, then every 3 months.

MANAGEMENT

- Discontinue the agent — functional features reverse in 8-12 wks; structural acromegaloid changes may not.
- Address comorbidities — hypertension, glucose intolerance, OSA; cardiology referral if LVH or symptoms.
- If IGF-1 stays elevated > 8-12 wks off therapy, proceed to full workup for endogenous GH excess (coincident acromegaly has been unmasked by secretagogue use).
- Counsel against resumption; document informed consent if the patient continues.

OVERDOSE ASSESSMENT

Stabilize & characterize: ABCs first (EMS for instability); intent (accidental vs self-harm); substance, formulation, and source; intended vs actual dose and units (mg/mcg/"units"); time course and stacking; patient factors; current symptoms by system.

No specific antidote exists — management is symptomatic and supportive.

ADVERSE EFFECTS AND OVERDOSE • CONTINUED**WORKUP, SYMPTOM-GUIDED**

- Electrolytes, creatinine, ALT/AST
- CK if muscle pain or dark urine
- Glucose for IGF-1/GH peptides
- C-peptide to separate exogenous insulin (suppressed) from endogenous hyperinsulinism (elevated)
- Troponin/ECG for chest pain
- Rule out co-ingestions (acetaminophen, salicylate, ethanol)

PEARLS

- "Any dose is an overdose" for most of these agents
- Always ask about injectable peptides (patients won't volunteer)
- Prolonged monitoring for long-acting agents (CJC-1295 t_{1/2} 5.8-8.1 d)
- Adolescent use risks premature epiphyseal closure
- Report reactions to Canada Vigilance / FDA MedWatch

The regulatory landscape

What makes a peptide a drug, the two compounding pathways, and why the label on the vial does not settle the question.

WHAT MAKES A PEPTIDE A DRUG

Regulatory status follows the article and its intended use, not the syringe. Under the FDCA a drug is any article intended to treat disease or to affect the structure or any function of the body, and no new drug may enter interstate commerce without an effective approved application. Intended use is read objectively — from labeling claims, advertising, statements, and the circumstances of distribution — so status does not change because a label disclaims the contemplated use.

Route enters later, and it matters. FDA's safety concerns about compounded peptides are route-specific: immunogenicity from aggregation, and difficulty characterizing peptide-related impurities. Route appears inside individual FDA entries for that reason — GHK-Cu is treated separately for injectable and non-injectable administration; GHRP-2 was flagged for injectable and nasal routes. It also determines whether sterility and cGMP obligations attach. What it does not do is convert a never-approved peptide into a lawful article: a topical preparation is an unapproved new drug as much as an injection is.

Off-label prescribing presupposes an approved product to be off the label of; for a never-approved peptide the marketing prohibition is unrelieved and the adequate-directions exemption is unavailable.

THE REGULATORY LANDSCAPE • CONTINUED

THE TWO COMPOUNDING PATHWAYS

503A (21 U.S.C. 353a)	Compounding by a licensed pharmacist in a state-licensed pharmacy or federal facility, or by a licensed physician, on a prescription for an individual patient. Compliance exempts the product from three sections only: cGMP, adequate directions for use, and new drug approval.
503B (21 U.S.C. 353b)	Compounding in a registered outsourcing facility, which need not be a licensed pharmacy and may compound without patient-specific prescriptions — what permits office stock. Its exemption omits cGMP.
Bulk drug substance	By regulation, the same as an active pharmaceutical ingredient.
Certificate of analysis	A statutory condition under 503A, alongside manufacture by an FDA-registered establishment. The statute sets no purity figure; 98% by HPLC is a purchasing norm, while compendial and cGMP standards fix purity, sterility and endotoxin limits.

Under 503A the three routes to bulk eligibility are hierarchical, not alternative. A substance must comply with an applicable USP or NF monograph where one exists. Only if no monograph exists may it qualify as a component of an FDA-approved drug. Only if neither is true does the third route open — appearance on the 503A bulks list. Three further conditions sit alongside: the substance cannot be on the withdrawn-or-removed list, the product cannot be a copy of a commercially available drug made in inordinate amounts, and it cannot be one FDA identifies as demonstrably difficult to compound.

THE BULKS LIST AND WHAT THE CATEGORIES ARE NOT

The 503A bulks list is codified at 21 C.F.R. 216.23 and holds six substances — Brilliant Blue G, cantharidin, diphenylcyclopropanone, N-acetyl-D-glucosamine, squaric acid dibutyl ester, thymol iodide — four topical-only. No peptide is on it. Listing happens by notice-and-comment rulemaking after FDA consults its advisory committee and balances four criteria: chemical characterization, safety, historical compounding use, and evidence of effectiveness. Even listing carries no endorsement — the regulation states the data are inadequate to establish general recognition of safety or effectiveness for any listed substance.

THE REGULATORY LANDSCAPE • CONTINUED

Categories 1, 2 and 3 are something else: enforcement-discretion postures described in FDA guidance while nominations are evaluated. Category 1 means FDA does not intend to act against a compounder using the substance if guidance conditions are met. Category 2 means FDA identified significant safety risks and will not extend that posture. Category 3 means the nomination lacked information FDA could evaluate. Substances nominated on or after January 7, 2025 are not categorized. Category status describes enforcement posture, not eligibility:

- As of FDA's May 2026 posting, 503A Category 2 holds six substances; the peptide-class entries are ibutamoren mesylate and kisspeptin-10.
- Most peptides once in Category 2 — BPC-157, TB-500, KPV, MOTS-c, DSIP, semax, epitalon, melanotan II, thymosin alpha-1, CJC-1295, AOD-9604, LL-37, dihexa, PEG-MGF, injectable GHK-Cu — now sit in a separate table of nominations withdrawn by their nominators. The safety write-ups remain posted; withdrawal is not clearance.
- GHK-Cu except for injectable routes returned to Category 1 in May 2026 after a nominator clarified a partial withdrawal; FDA intends to consult its advisory committee on GHK-Cu before the end of February 2027.
- At the July 2026 advisory committee meeting FDA reviewed seven peptides against narrow named indications — BPC-157 for ulcerative colitis, TB-500 and KPV for wound healing, epitalon for insomnia. A favorable recommendation still leaves rulemaking ahead of listing.

THE RESEARCH-USE SUPPLY CHANNEL

Two provisions get invoked as though they authorized human use. Neither does. "For Research Use Only. Not for use in diagnostic procedures." comes from 21 C.F.R. 809.10(c), a labeling provision for in vitro diagnostic products in the laboratory research phase; its effect is to excuse a shipment from IVD labeling requirements. The drug-side analogue, 21 C.F.R. 312.160, permits shipping an investigational drug without an IND for laboratory animal or in vitro use if the label reads "Not for use in humans" — and obliges the shipper to confirm the consignee runs such tests. FDA may terminate that authorization where the drug goes to anything but bona fide investigation.

Imported material fares no better. An article is refused admission if it is adulterated, misbranded, or in violation of the approval requirement, and FDA states that personal importation is in most circumstances illegal. The discretion FDA describes is conditioned on facts this market does not present: no promotion to US residents, generally not more than a three-month supply, and a named US-licensed treating physician.

THE REGULATORY LANDSCAPE • CONTINUED

PRESCRIBER-FACING EXPOSURE

Exposure runs on three independent tracks. Federal: introducing, receiving or delivering an adulterated or misbranded drug in interstate commerce are prohibited acts, punishable by up to a year's imprisonment without proof of intent, rising to three years with a prior conviction or intent to defraud. State: board authority rests on statutory definitions of unprofessional conduct reaching gross negligence, repeated negligent acts, incompetence, and assisting a violation of the practice act — none requiring that the prescriber handled a product, and all reaching a referral to a vendor. Civil standard-of-care and causation questions are treated at length in The Expert Witness Pocket Guide.

1. Document the identity of the article and its regulatory status in the chart, in those terms — not as off-label use of an approved drug.
2. Where a compounded preparation is used, record the pathway (503A or 503B), the facility, and the certificate of analysis relied on.
3. Build consent around the elements FDA's research-consent regulation enumerates: that the therapy is not approved for the indication, the foreseeable risks, the approved alternatives, and what is available if injury occurs.
4. Record the clinical reasoning and monitoring plan; a board reviews reasoning, not footnotes.
5. Verify category and listing status at the time of prescribing — these lists changed materially in April and May 2026 alone.

The 503A bulks list and the July 2026 committee

The 503A bulks list as it actually stands, the July 2026 advisory committee, and what a committee recommendation is worth.

PEPTIDES AND THE SECTION 503A BULKS LIST: STATUS AS OF AUGUST 2026

Under section 503A of the Federal Food, Drug, and Cosmetic Act, a bulk drug substance may be used in compounding only if it meets a USP or NF monograph, is a component of an FDA-approved drug, or appears on FDA's 503A Bulks List. Most research peptides satisfy none of the first two, so the Bulks List decides whether they can lawfully be compounded at all.

FDA's Pharmacy Compounding Advisory Committee met on July 23-24, 2026 under Docket No. FDA-2025-N-6895, announced at 91 Fed. Reg. 20465 (Apr. 16, 2026). Seven peptide families were considered, each in free base and acetate form, for a total of fourteen individual substances put to separate votes:

- BPC-157 – reviewed for ulcerative colitis
- KPV – reviewed for wound healing and inflammatory conditions
- TB-500 – reviewed for wound healing
- MOTS-c – reviewed for obesity and osteoporosis
- Emideltide (delta sleep-inducing peptide, DSIP) – reviewed for opioid withdrawal, chronic insomnia, and narcolepsy
- Semax – reviewed for cerebral ischemia, migraine, and trigeminal neuralgia
- Epitalon – reviewed for insomnia

FDA review staff recommended against listing all fourteen. Each briefing document concludes that a balancing of the four statutory criteria weighs against placing the substance on the list. The grounds were consistent: the substances are not well characterized chemically, nominations lacked certificates of analysis and impurity, aggregate or endotoxin controls, injected peptides carry immunogenicity risk, and approved drugs already exist for every condition reviewed.

THE 503A BULKS LIST AND THE JULY 2026 COMMITTEE • CONTINUED

What the committee voted is not documented. FDA posted briefing documents, presentations, the agenda, the roster and the fourteen voting questions — but no minutes, transcript or tally. Vote counts appearing in trade press cannot be confirmed against FDA's materials and should not be relied on.

WHY THIS IS NOT YET A FINAL ANSWER

A committee recommendation is advisory. FDA is not bound by it, and the agency's own briefing document states that it does not intend to issue a final determination until the advisory committee input has been considered and all reviews are finalized. A substance joins or is formally excluded from the 503A Bulks List only when FDA amends 21 C.F.R. 216.23 by rulemaking.

As of August 2026 no such rule has issued. Section 216.23 still lists only six permitted substances and four excluded ones, and none of these seven peptides appears in either paragraph. The matter is unresolved.

Practical effect: none of these seven peptides is on the 503A Bulks List, so compounding them under section 503A is not authorized by that route — and this was already true before the July 2026 meeting, which changed nothing legally.

503A Bulks List

Substances FDA permits in section 503A compounding, codified at 21 C.F.R. 216.23(a).

PCAC

Advisory committee that reviews nominated substances; its recommendations are non-binding.

Advisory recommendation

Committee or staff advice. Carries no legal force on its own.

Final rule

The only mechanism that adds a substance to, or formally excludes it from, the list.

Current status

Not listed, not formally excluded, no final rule pending publication.

Cosmetic, compounded, or supplement?

Three regulatory regimes reaching the same molecule, and what each permits.

The same peptide can be sold under three very different regulatory regimes depending on its route and the claims made — not its molecular identity. The regime, not the molecule, determines what is lawful.

COSMETIC — TOPICAL, APPEARANCE CLAIMS ONLY

Under the FD&C Act and the Modernization of Cosmetics Regulation Act (MoCRA, 2022), a topical product that makes only appearance claims (moisturizing, improving the look of fine lines) is a cosmetic; no premarket approval, but safety and labeling rules apply. A therapeutic claim (healing, regenerating tissue, treating a condition) reclassifies it as an unapproved drug.

Example: a GHK-copper cream sold for skin appearance is a cosmetic; the identical cream marketed to "regenerate tissue" crosses into drug territory.

COMPOUNDED — PRESCRIPTION, 503A / 503B

A licensed pharmacy may compound a peptide for an individual patient prescription (503A) or as cGMP batches for clinics (503B). The bulk substance must ordinarily satisfy a statutory pathway: an applicable USP or NF monograph, a component of an FDA-approved drug, or the 503A Bulks List itself. Separately from those routes, FDA's interim policy describes circumstances in which the agency does not intend to act on certain Category 1 nominated substances while evaluation is pending — enforcement posture, not statutory eligibility. A COA for identity, purity and sterility is expected either way.

Example: GHK-copper sits in Category 1 for non-injectable routes, so FDA does not currently intend to act against its use in compounding pending evaluation. That is not the same as statutory eligibility, and most injectable research peptides satisfy no pathway at all.

COSMETIC, COMPOUNDED, OR SUPPLEMENT? • CONTINUED**DIETARY SUPPLEMENT — ORAL, DSHEA**

Marketing a peptide as an oral "dietary supplement" under DSHEA is the most legally contested route. Most synthetic peptides do not meet the statutory definition of a dietary ingredient and lack a New Dietary Ingredient (NDI) notification; the FDA has stated that several — including BPC-157 — are excluded from the supplement category and treated as unapproved drugs.

Example: oral BPC-157 capsules are widely sold as "supplements," but the FDA has issued warning letters rejecting that classification.

Bottom line: route plus claim define the regime — the same peptide can be a lawful cosmetic, a prescription compound, or an unlawful "supplement" depending on how it is sold.

Jurisdictional snapshot

How the same molecule is scheduled differently across national regulators.

Australia (TGA)	BPC-157 Schedule 4 + Appendix D — possession without authority is a criminal offence.
UK (MHRA)	not licensed for human use.
EU	no EMA authorization; member-state enforcement.
New Zealand	prescription medicine since 2023.
UAE/GCC	Federal Decree-Law 38 (2024) permits licensed compounding under valid prescriptions to GCP/USP 795/797.
WADA	Section S0 prohibits all non-approved substances; no TUE.

Research-use-only (RUO) products carry most gray-market trade. The "not for human consumption" disclaimer offers no legal protection — the FDA regulates by intended use, inferred from marketing, dosing calculators, and reviews (FDA has issued warning letters to telehealth platforms and online sellers of compounded GLP-1 products; check the agency's warning-letter database for the current set rather than relying on a circulated count).

Independent assays of illegally-purchased material have found content well outside label and purity far below claim, alongside endotoxin, residual TFA, and heavy-metal contamination. No single published range describes the market as a whole.

JURISDICTIONAL SNAPSHOT • CONTINUED**TWO LEGAL PITFALLS FOR PROVIDERS**

"Off-label" doesn't apply to unapproved drugs. Off-label use is a legal pathway only for drugs holding at least one FDA-approved indication. Prescribing BPC-157 or any never-approved peptide is use of an unapproved new drug — IND, Expanded Access, and Right-to-Try do not cover routine wellness use.

"Recommending" is not a liability shield: directing patients to source research-grade peptides for self-administration is practicing medicine; liability and board jurisdiction attach regardless of whether the practitioner handled the product, and most malpractice policies exclude unapproved-drug claims.

Bottom line: if a peptide is not FDA-approved, you are operating outside the drug-approval framework — document the rationale, product source, and informed consent.

Case study: Lilly v. FDA

Retatrutide and the 40-amino-acid boundary, from complaint to pending appeal.

Retatrutide and the >40-amino-acid boundary (2023–2025)

THE MOLECULE AND THE DISPUTE

Retatrutide (Lilly's investigational GLP-1/GIP/glucagon triple agonist) has a 39-alpha-amino-acid backbone plus a covalent side chain of one gamma-glutamate and one ADO residue — the latter not an alpha amino acid. Counted cumulatively it has 41 residues; counted as alpha amino acids, 39 (sequence patent-disclosed: US20230143781A1).

In November 2023 Lilly asked the FDA to classify it as a biological "protein" under the BPCIA's >40-amino-acid threshold, counting all 41 residues. The FDA counted only the 39 backbone alpha amino acids, excluded the ADO residue, and ruled it a drug under the FD&C Act — declining to designate it "analogous to a protein."

THE RULING AND WHY IT MATTERS

Lilly sued (*Eli Lilly & Co. v. Kennedy*, No. 1:24-cv-01503-TWP-KMB, S.D. Ind.). The September 30, 2025 order split: the court upheld FDA's determination that retatrutide does not meet the regulatory definition of a protein, but vacated FDA's separate determination that it could not be "analogous to a protein" and remanded that question to the agency. Final judgment entered December 16, 2025. It did not establish that retatrutide is a biologic.

Lilly appealed in February 2026 (No. 26-1301, 7th Cir.), briefed through June 2026, and the appellate proceeding remains pending as of 13 August 2026. As a single district-court ruling under appeal it is persuasive rather than settled national law, but it is the cleanest case to date on how the >40-residue threshold pivots on which residues you count — with large commercial stakes: BLA vs NDA pathways, 12-year biologic vs shorter NCE exclusivity, biosimilar vs generic competition, and very different 503A compounding posture.

CASE STUDY: LILLY V. FDA • CONTINUED**CLINICAL RELEVANCE FOR PROVIDERS**

For prescribers and compounding pharmacies, classification as a biologic would put generic competition under the BPCIA rather than the FD&C Act — shifting the market-exclusivity and biosimilar timeline — and would change whether and how retatrutide could ever enter a compounding pathway, since 503A compounding of biologics operates under different authority than for small-molecule drugs.

Classification is not finally settled: the court upheld FDA's “protein” determination but vacated and remanded the separate “analogous to a protein” determination, so retatrutide was not established as a biologic and its status remains pending on appeal.

Case citation

Eli Lilly & Co. v. Kennedy, No. 1:24-cv-01503-TWP-KMB (S.D. Ind.) (complaint filed 3 Sep. 2024; partial summary judgment 30 Sep. 2025; final judgment 16 Dec. 2025), appeal pending, No. 26-1301 (7th Cir.)

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